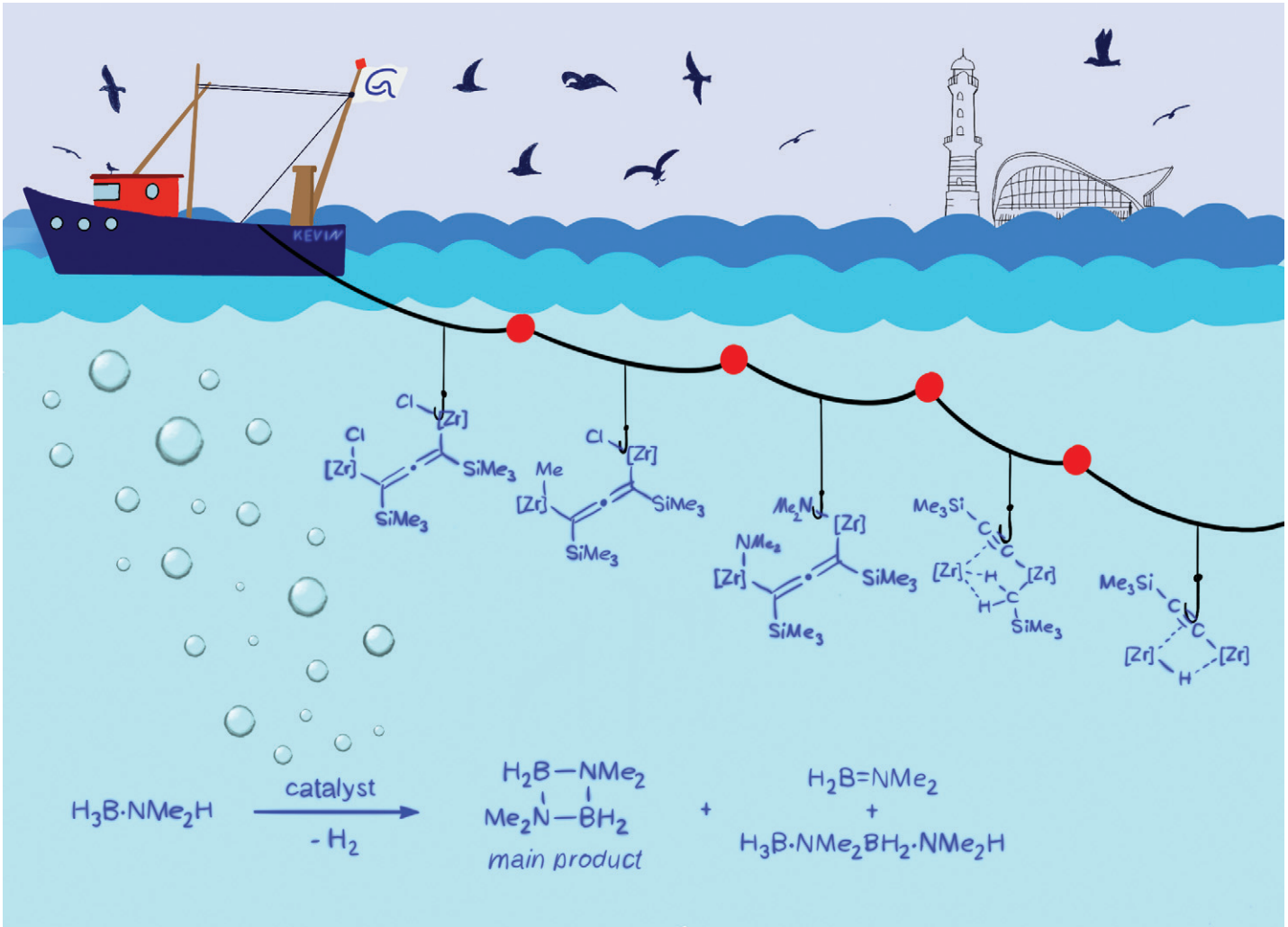


Showcasing research from the Beweries labs at the Leibniz Institute for Catalysis (LIKAT Rostock), Germany.

Mechanistic insights into dehydrocoupling of amine boranes using dinuclear zirconocene complexes

Catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$ using the in situ system $Cp_2Zr(Cl)(\mu\text{-}Me_3SiC_3SiMe_3)Zr(Cl)Cp_2$ (**1**)/MeLi was studied as a model for dehydropolymerisation of $H_3B{\cdot}NMeH_2$. Spectroscopic monitoring of the reaction as well as a series of stoichiometric experiments show the formation and interconversion of several catalytically active zirconocene species.

As featured in:



See Fabian Reiß, Torsten Beweries *et al.*, *Catal. Sci. Technol.* , 2021, **11**, 4034.

ROYAL SOCIETY OF CHEMISTRY

rsc.li/catalysis

Registered charity number: 207890

Catalysis Science & Technology

ROYAL SOCIETY OF CHEMISTRY

PAPER

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Catal. Sci. Technol.*, 2021, 11, 4034

Mechanistic insights into dehydrocoupling of amine boranes using dinuclear zirconocene complexes†

Kevin Lindenau,[a] Nora Jannsen,[a] Mirko Rippke,[a] Hanan Al Hamwi,[a] Carmen Selle,[a] Hans-Joachim Drexler,[a] Anke Spannenberg,[a] Mathias Sawall,[b] Klaus Neymeyr, [ab] Detlef Heller,[a] Fabian Reiß *[a] and Torsten Beweries *[a]

Received 26th March 2021,
Accepted 28th April 2021

DOI: 10.1039/d1cy00531f

rsc.li/catalysis

Catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$ using the *in situ* system $Cp_2Zr(Cl)(\mu\text{-}Me_3SiC_3SiMe_3)Zr(Cl)Cp_2$ (**1**)/MeLi was studied as a model for previously reported dehydropolymerisation of $H_3B{\cdot}NMeH_2$. NMR and UV-vis spectroscopic monitoring of the precatalyst activation sequence as well as a series of stoichiometric experiments showed that formation of a zirconocene dimethyl complex (**2**) is not relevant for activation of the precatalyst. Instead, deprotonation of $H_3B{\cdot}NMe_2H$ and reaction of thus formed $Li[NMe_2BH_3]$ is proposed to result in the formation of Zr amidoborane and hydride species. DFT analysis using such complexes as active species showed a pathway for formation of H_2 and $H_2B{=}NMe_2$. ^1H NMR spectroscopic monitoring and stoichiometric control experiments revealed the formation of unusual diamagnetic dinuclear complexes $Cp_2Zr(C_2SiMe_3)(\mu\text{-}R)ZrCp_2$ (R = CH_2SiMe_3, **7**; R = H, **9**) formed by activation of the allenediide unit of the precatalyst **1**. Such species can be regarded as rare single-component catalysts for the dehydrocoupling of amine boranes.

Introduction

The atom-efficient catalytic formation of bonds between main group elements such as boron and nitrogen is of great interest in the fields of organometallic and materials chemistry as it could provide an entry point to new main group compounds and materials.[1] Up to now, a plethora of early and late transition metal complexes are known to serve as catalysts for the dehydrocoupling of primary and secondary amine boranes.[2] Coordinatively and electronically unsaturated early transition metal complexes, especially those of group 4 metals are very interesting in this context as these are known to possess unique, high reactivities with N–H and B–H bonds.[3] Examples for the use of group 4 metallocene systems that constitute metallocene dihalide and organolithium reagents were reported in the past,[4] however, in these systems the elucidation of the mechanistic details is complicated by the presence of the strong base. Also, single-component systems based on titanocene(III) amidoborane, hydride, and alkyl complexes are known.[5] We have shown that titanocene and zirconocene η^2-bis(trimethylsilyl)acetylene complexes are potent precatalysts for dehydrocoupling of dimethylamine borane, $H_3B{\cdot}NMe_2H$, and hydrazine borane, $H_3B{\cdot}N_2H_4$, that generate the reactive 14 electron $[Cp_2M]$ species at room temperature without the need for strong base.[6] Ti based catalyst systems were generally found to be more active than their Zr analogues. While the role of M(III) intermediates was not evaluated in these studies, previous reports indicate that especially Ti complexes in the oxidation state +3 can be interesting for dehydrocoupling reactions.[5,7]

Recently, we have reported on a unique type of dinuclear zirconocene allenediide bridged precatalyst (**1**) for dehydropolymerisation of methylamine borane, $H_3B{\cdot}NMeH_2$ (Scheme 1a).[8] After activation with methyl lithium (MeLi) this precatalyst produces comparably low-molecular weight poly(aminoborane)s, presumably through an on-metal pathway according to the spectroscopic data and analysis of molecular weights at different reaction conditions. A dinuclear zirconocene methyl complex (**2**) with intact allenediide bridging unit was proposed as an intermediate that could upon reaction with $H_3B{\cdot}NMeH_2$ produce spectroscopically observed methane and a putative zirconocene bis(amidoborane) species **3a** (Scheme 1a). NMR spectroscopic analysis of catalytic mixtures suggested the presence of several unidentified hydride species, possibly including the dihydride **ZrH–ZrH**. However, the exact nature

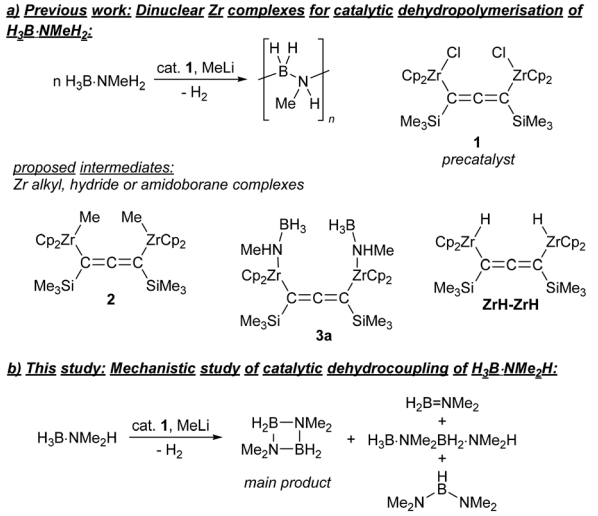
[a] *Leibniz-Institut für Katalyse e.V, Albert-Einstein-Str. 29a, 18059 Rostock, Germany. E-mail: fabian.reiss@catalysis.de, torsten.beweries@catalysis.de*
[b] *Institut für Mathematik, Universität Rostock, Universitätsplatz 1, 18055 Rostock, Germany*

† Electronic supplementary information (ESI) available: CCDC 2046121, 2046122, 2069777 and 2069778. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/d1cy00531f

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

This journal is © The Royal Society of Chemistry 2021

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.



Scheme 1 A dinuclear zirconocene allenediide bridged precatalyst (**1**) for dehydrocoupling of amine boranes.

of the subsequent elementary steps that lead to catalytic turnover remained unclear. Dinuclear complexes of late transition metals were reported to be of relevance for catalytic dehydrocoupling of amine boranes, *e.g.* by the Weller group for the case of Rh diphosphine complexes.[9]

Reactions with the higher substituted analogue dimethylamine borane, $H_3B{\cdot}NMe_2H$, can give valuable insights into the reaction mechanisms of amine borane dehydrocoupling[10] as this substrate shows a higher solubility, all intermediates and reaction products are soluble in non-polar solvents and release of more than one equivalent of hydrogen is prevented by the presence of an additional methyl group. As the previously reported dinuclear complex **1** shows exceptional catalytic activity for a Zr based dehydrocoupling catalyst, especially in comparison to mononuclear systems, we have now performed a study of dehydrocoupling of $H_3B{\cdot}NMe_2H$ using the same precatalyst with MeLi as the cocatalyst. In this paper, we systematically elucidate the precatalyst activation sequence, the speciation of the catalyst, and the role of dinuclear Zr(III) complexes in this reaction.

Results and discussion

Catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$

Catalytic studies. Dehydrocoupling of $H_3B{\cdot}NMe_2H$ was studied using the same protocol as used before for dehydropolymerisation of $H_3B{\cdot}NMeH_2$,[8] *i.e.* addition of two equivalents of MeLi to a toluene solution containing the precatalyst **1** (2.5 mol%) and the amine borane ($[H_3B{\cdot}NMe_2H]$ = 0.33 M) with volumetric monitoring of the reaction system (Scheme 1b).[11] Gas evolution started immediately after addition of MeLi and a colour change from orange to red was observed. With time the colour of the reaction mixture changed to green and finally to colourless at maximum conversion of the substrate (see ESI† for details, Fig. S2). Notably, the same colour gradient was observed by us before for the $H_3B{\cdot}NMeH_2$ system.[8] According to the amount of evolved gas, no full conversion of $H_3B{\cdot}NMe_2H$ could be achieved ($n(H_2)/n(H_3B{\cdot}NMe_2H)$ = 0.85) (Fig. 1, top). Variations of the catalyst concentration showed that similar degrees of conversion of the substrate can also be observed at lower [**1**], however, reactions were much slower in these cases. To our surprise, runs with 0.63 and 1.25 mol% **1** reproducibly showed very similar volumetric traces. Also, reactions with 3.75 or 5 mol% **1** were much slower than those with 2.0 or 2.5 mol% **1**. A rather complex mechanism of catalyst activation, dehydrogenation of amine borane, deactivation, and reactivation of the catalytically active species that possibly involves the presence of multiple Zr species is thus expected.

Based on the set of volumetric plots obtained from variation of the precatalyst concentration, no kinetic analysis (*e.g.* time normalisation analysis[12]) was possible to determine the order in **1**. Similarly, reactions with varying $[H_3B{\cdot}NMe_2H]$ reproducibly showed drastically different kinetic profiles (Fig. S3†) that prevent further analysis. It should however be noted that all volumetric profiles systematically show regimes of different activity. Within the first hour very fast H_2 evolution

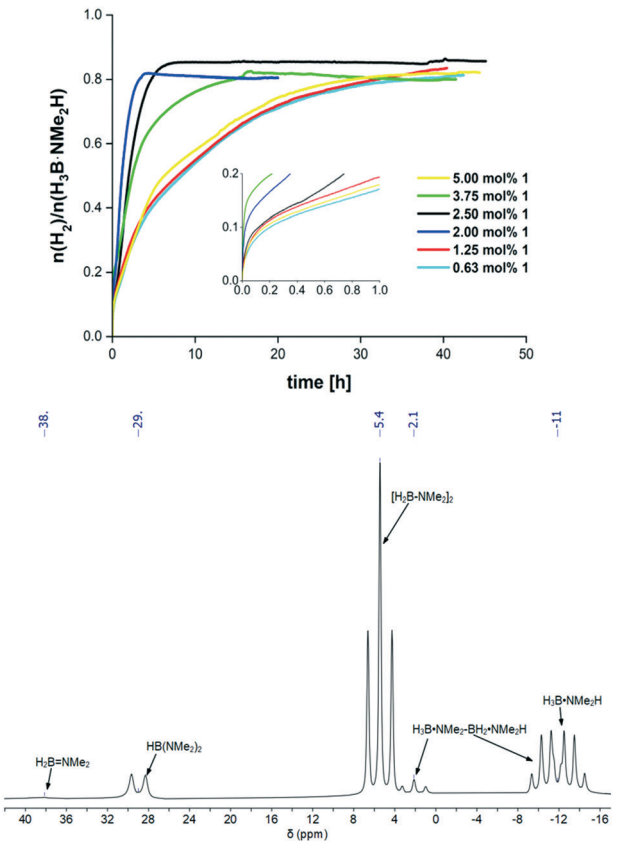


Fig. 1 Top: volumetric curves of dehydrogenation of $H_3B{\cdot}NMe_2H$ using *in situ* mixtures of complex **1**/MeLi (toluene, 25 °C). Bottom: ^{11}B NMR spectra after dehydrocoupling of $H_3B{\cdot}NMe_2H$ using 2.5 mol% of **1** and 5.2 mol% MeLi (297 K, toluene-d_8, 96.7 MHz).

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

is observed, followed by a regime where H_2 formation flattens out towards the end of the reaction (Fig. 1).[13]

In reactions with deuterated substrates H_2 evolution traces with $H_3B{\cdot}NMe_2H$ and $D_3B{\cdot}NMe_2H$ are virtually identical, albeit with slightly lower initial activity for $D_3B{\cdot}NMe_2H$. Use of both B-deuterated substrates $D_3B{\cdot}NMe_2H$ and $D_3B{\cdot}NMe_2D$ gave a reduction of the initial rate of the reaction, suggesting that B–H activation could be relevant to slow formation of a catalytically active species. N-Deuterated substrates $H_3B{\cdot}NMe_2D$ and $D_3B{\cdot}NMe_2D$ display a similar behaviour at the onset of the reaction followed by a significant decrease in activity at higher conversions with linear H_2 evolution (Fig. S5 and S6†), pointing to a different turnover-limiting step for these isotopomers. We thus refrain from a discussion of kinetic isotope effects for this system. However, it must be assumed that similar as for other early transition metal-based systems,[4b,8] cleavage of the N–H bond is more relevant to the rate determining steps of catalysis than B–H activation. It should additionally be noted that especially for early transition metal-based systems, interpretation of KIE experiments with N-deuterated amine boranes can be complicated by catalyst deactivation by traces of water that could be present from the deuteration of the substrate.[5c]

In line with the volumetric profiles, ^{11}B NMR spectroscopic analysis of the reaction solution revealed the presence of considerable amounts of residual amine borane starting material. The product distribution that is observed under catalytic conditions resembles those reported before[14] with the cyclic diborazane, $(H_2BNMe_2)_2$, as the main product (δ 5.4 ppm, d, J_{BH} = 113.0 Hz) and linear diborazane, $H_3B{\cdot}NMe_2BH_2{\cdot}NMe_2H$ (δ −11.2 ppm, q, J_{BH} = 93.7 Hz; 2.1 ppm, t, J_{BH} = 108.7 Hz) as well as diaminoborane, $HB(NMe_2)_2$, as minor components (δ 29.7 ppm, d, J_{BH} = 135.5 Hz) (Fig. 1, bottom). Close inspection of the high-field region of ^{11}B NMR spectra indicated the presence of μ-aminodiborane, $BH_2(\mu\text{-}Me_2N)(\mu\text{-}H)BH_2$ (δ −20.9 ppm, q, J_{BH} = 89.7 Hz). This species could form upon B–N cleavage in the amine borane and release of amine[15] (Fig. S45†), which could potentially interact with the catalyst or intermediates thereof. This suggestion is supported by a two-dimensional potential energy surface scan (2D-PES) where we could identify two possible reaction paths for the catalyst-free formation of linear diborazane from aminoborane and amine borane (Fig. S101†).[16] Both pathways show first the B–N cleavage in amine borane followed by stepwise addition of borane to aminoborane. Furthermore, this 2D-PES clearly shows the exergonic formation of μ-aminodiborane and free amine as another viable reaction path.

While the presence of free amine or even doping with excess amine was found to have a beneficial effect in previous studies using late transition metal-based systems,[17] protonation of alkyl, allenediide or hydride ligands and formation of stable Zr-amide linkages[18] could negatively affect the catalytic activity of the herein described Zr based system. Addition of an equimolar amount of NMe_2H during catalysis results in inhibition of dehydrocoupling with reactions giving

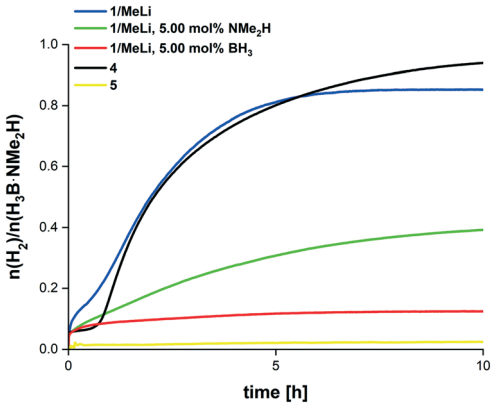
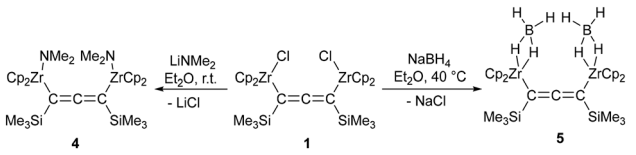


Fig. 2 Comparison of the volumetric curves of dehydrocoupling of $H_3B{\cdot}NMe_2H$ using **1**/MeLi, potential catalyst poisons, and isolated amide (**4**) and borate complexes (**5**) (2.5 mol% precatalyst, toluene, 25 °C).

only approximately 0.4 equivalents of H_2 after 10 hours (Fig. 2).[19] To evaluate the assumed formation of Zr amide species, we have next performed an experiment using the dinuclear Zr bis(dimethylamide) complex $Cp_2Zr(NMe_2)(\mu\text{-}Me_3SiC_3SiMe_3)Zr(NMe_2)Cp_2$ (**4**), prepared from reaction of **1** with $LiNMe_2$ (Scheme 2). ^1H NMR spectra of this species showed the expected Cp and $SiMe_3$ resonances as well as a characteristic resonance for the NMe_2 group (δ 2.86 ppm).[20] To our surprise, this complex shows a pronounced induction period and full conversion of the substrate, *i.e.* formation of approximately 1.0 equivalent of H_2. NMR spectroscopic monitoring of stoichiometric and catalytic reactions of this complex with $H_3B{\cdot}NMe_2H$ (Fig. S52 and S73†) showed formation of $(H_2BNMe_2)_2$ as the main species and $HB(NMe_2)_2$ and $H_3B{\cdot}NMe_2BH_2{\cdot}NMe_2H$ as minor products. Addition of BH_3 (formed from B–N cleavage in $H_3B{\cdot}NMe_2H$) to the amide group of **4** could result in the formation of an amidoborane complex, which could be further converted into Zr alkyl and hydride species (*vide infra*). Nevertheless, we could not detect zirconocene species that are characteristic for the parent system **1**/MeLi. It can thus be assumed that this dinuclear amide complex **4**, which could form upon reaction of free amine with a postulated Zr methyl intermediate **2**, is not relevant to the dehydrogenation of $H_3B{\cdot}NMe_2H$ using the parent system **1**/MeLi. We therefore refrain from a more detailed evaluation of this species in this study. The evaluation of the potential of **4** for related dehydrocoupling experiments is currently ongoing in our group.

Similarly, we have evaluated the effect of BH_3 doping of the system **1**/MeLi (Fig. 2). Addition of 5 mol% $BH_3{\cdot}THF$ results in almost complete loss of activity. A Zr borate



Scheme 2 Synthesis of Zr amide complex **4** and borate complex **5**.

This journal is © The Royal Society of Chemistry 2021

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

complex could form by addition of BH_3 to a transient species with terminal hydride ligands (**ZrH–ZrH**). We have therefore prepared such a borate species **5** by salt metathesis of **1** and $NaBH_4$ (Scheme 2).[21] ^{1}H NMR spectra showed the expected Cp and $SiMe_3$ resonances, ^{11}B decoupling additionally produced a characteristic broad signal at 0.69 ppm that corresponds to the highly fluxional BH_4 groups.[22] Complex **5** shows no activity for dehydrogenation of $H_3B{\cdot}NMe_2H$. Based on these experiments we conclude that either free amine or borane, formed by B–N cleavage in $H_3B{\cdot}NMe_2H$ could serve as a catalyst poison for the system **1**/MeLi.

Mechanistic investigations

Activation of the precatalyst with MeLi. In our previous study, we had proposed formation of a dinuclear Zr dimethyl complex **2** by addition of MeLi to the precatalyst **1**.[8] We have now evaluated formation of this species more closely by *in situ* NMR and UV-vis spectroscopy. Addition of two equivalents of MeLi to the precatalyst in toluene-d_8 results in rapid formation of the proposed dimethyl complex **2** (^{1}H: δ 5.90, 5.92 ppm). UV-vis spectroscopic monitoring of the intensely coloured precatalyst activation sequence (*vide supra*) was done using a dip optrode[23] that was introduced into the reaction vessel. Complex **1** shows a pronounced UV-vis band at 400 nm (Fig. S76†) which upon addition of two equivalents of MeLi decays immediately (Fig. 3, top). Since the spectrum of the starting material **1** cannot be reconstructed after the addition of MeLi, it can be assumed that one or more intermediate products are already present at the beginning of the reaction. Within one hour, the evolution of bands (λ_{max} = 352 nm, 424 nm; λ_{min} = 335 nm) takes place, which can be assigned to the formation of new Zr species. When assuming that at full conversion the previously proposed complex **2** is present as the only product, it is reasonable to assign the second of these UV-vis patterns to the formation of a singly methylated complex **6**.

Additionally, analysis of the singular values (see Experimental section for details, Fig. S82†) revealed that more than two UV-vis active species are involved in this activation of **1**. It is known from the literature that organolithium compounds are capable of aggregating, both with themselves and with lithium halides.[24] To investigate a possible aggregation of MeLi itself under the chosen experimental conditions, it was stepwise diluted in toluene and monitored by UV-vis spectroscopy (Fig. S80†). If the extinction of different wavelengths is plotted against concentration of MeLi, no linear relationship is observed – the MeLi system thus does not conform to Beer's law (Fig. S80†). Moreover, a singular value decomposition of this titration showed already at least two UV-vis active species (Fig. S81†). Considering the concentration-dependent different options for aggregation of MeLi and the additional clustering with lithium halides, a multitude of possible reactions result. It must thus be concluded that for the herein described system operando UV/vis spectroscopy cannot be used to access reliable kinetic data, especially for the subsequent catalytic dehydrocoupling reaction.

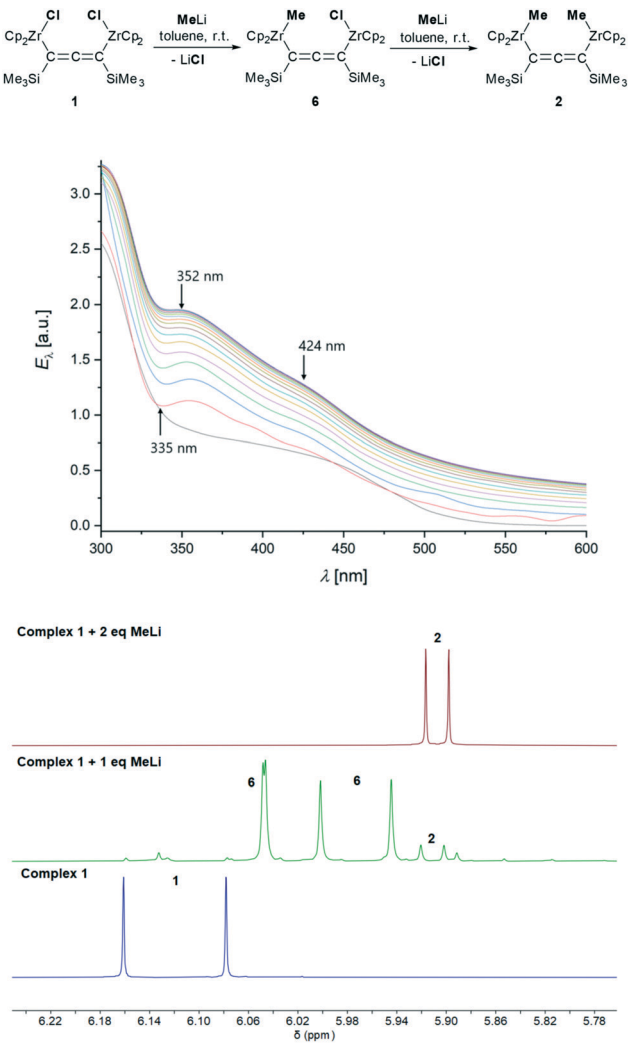


Fig. 3 Top: UV-vis reaction spectra for the of addition of two equivalents of MeLi to complex **1** in toluene at 25 °C ([**1**] = 3.1×10^{-4} M, [MeLi] = 6.3×10^{-4} M, reaction time 60 min, timeline shown from bottom to top). Bottom: comparison of ^{1}H NMR experiments of MeLi activation. Conditions: room temperature, benzene, [**1**] = 3.3×10^{-2} M, [MeLi] = 3.3×10^{-2} M (green) and [**1**] = 6.6×10^{-2} M, [MeLi] = 1.5×10^{-1} M (red).

The UV-vis spectroscopic detection of a new species **6** during activation of the precatalyst under dilute conditions contrasts the above-described findings from an NMR scale activation reaction in more concentrated solution where **2** was obtained exclusively. We assign this to differences in concentration of **1** that should affect the amounts of **2** and **6** present in the equilibrium mixture of the consecutive reaction (Fig. 3, top). To corroborate this, we have repeated the ^{1}H NMR experiment using only one equivalent of MeLi for activation of **1** and found that indeed transformation of **1** into **2** proceeds *via* the above-mentioned new species **6** that shows characteristic Cp resonances of equal intensity at δ 6.05, 6.05, 6.00, and 5.94 ppm. Under these conditions, this

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

complex is the main species. Single crystals of **6** suitable for an X-ray analysis could be isolated from a larger scale reaction at T = 40 °C after crystallisation from diethyl ether. The molecular structure (Fig. 4) confirms the presence of a singly methylated species, albeit with compound **2** also being part of the asymmetric unit.

Notably, three isomers of **6** are present in the solid state that differ in the orientation of the methyl and chlorido ligand at the Zr centre with respect to the allenediide unit (*i.e.* in_{Cl}–in_{Me} out_{Cl}–in_{Me} and in_{Cl}–out_{Me} isomer). This observation is well in line with our calculations were the in_{Cl}–in_{Me} isomer was found to be thermodynamically preferred (Gibbs free energies related to the in_{Cl}–in_{Me} isomer $\Delta_R G^{\theta}$ = 2.4 (in_{Cl}–out_{Me}), 2.7 (in_{Me}–out_{Cl}), 4.3 (out_{Cl}–out_{Me}) kcal mol^{-1}). Although both molecules in the asymmetric unit show a Cl/Me disorder, the content of **2** can be determined in combination with NMR spectroscopy. In ^{1}H NMR measurement we identified only trace amounts of **1** in the isolated crystals (Fig. S19†). With this knowledge we neglected the amount of **1** and calculated a crystallographic content of 11% of species **2**. Furthermore, the calculated distribution of in_{Cl}–in_{Me} *vs.* in_{Cl}–out_{Me} *vs.* out_{Cl}–in_{Me} isomers of 48/34/18% shows good agreement with the calculated thermodynamic values.

Most interestingly, the described reaction sequence requires a significant amount of time for completion, which in turn could explain the observed varying activities at the onset of catalytic dehydrocoupling reactions (Fig. S1†). Addition of alkyl lithium reagents to metallocene halides is a commonly applied procedure for *in situ* precatalyst activation in early transition metal chemistry and formation of the respective alkyl complexes is typically assumed to be quantitative within short reaction times.[4,26] We would like to point out that such protocols could be problematic in cases where quantitative kinetic interpretations are done.

Comparison with other precatalyst activation protocols. With this information on the complexity of the methylation of **1** in hand we next evaluated alternative protocols for the generation of the active species. As mentioned above, addition of MeLi to the precatalyst **1** results in approximately 85% conversion of $H_3B{\cdot}NMe_2H$ after 6 hours. Use of the singly methylated species **6** results in similar volumetric profiles (Fig. 5, top), suggesting that such species could also be relevant for activation of the precatalyst and generation of a proposed hydride species. In our previous study, we had presented the dimethyl complex **2** as an intermediate present during dehydropolymerisation of $H_3B{\cdot}NMeH_2$. We have now optimised the protocol for the synthesis and isolation of this species and used it as a precatalyst for dehydrocoupling of $H_3B{\cdot}NMe_2H$. Unexpectedly, reactions using this complex showed full conversion but were considerably slower compared to the parent system **1**/MeLi (Fig. 5, top). This observation could be assigned to the highly sensitive nature of this dinuclear dimethyl complex, which decomposes in solution at room temperature (Fig. S10†). We have therefore also prepared complex **2** *in situ* by addition of MeLi to the precatalyst, followed by addition of the substrate after an activation time of one hour. However, this reaction produced much less H_2. Alternatively, initial deprotonation of the substrate with MeLi to form $Li[NMe_2BH_3]$, followed by abstraction of the chloride ligand from the precatalyst could be a viable option. We have thus tested whether $Li[NMe_2BH_3]$[27] can be used as a cocatalyst instead of MeLi and found that in this case similar conversion of $H_3B{\cdot}NMe_2H$ as for the **1**/MeLi system is observed. However, a significant induction

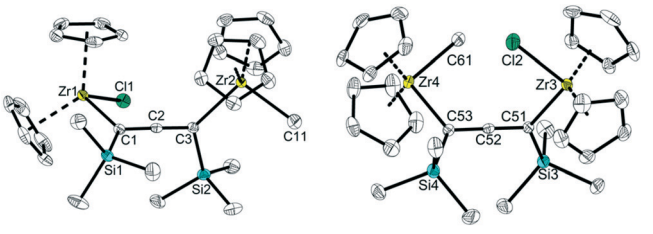


Fig. 4 Molecular structure of both molecules of the asymmetric unit of complex **6**. Left: in_{Cl}–out_{Me} isomer. Right: in_{Cl}–in_{Me} isomer. The crystals contain complexes **6** and **2** in 89 : 11 ratio, the latter species and the out_{Cl}–in_{Me} isomer have been omitted for clarity. Thermal ellipsoids correspond to 30% probability. Minor occupied chlorido and methyl ligands as well as hydrogen atoms are omitted for clarity.[25]

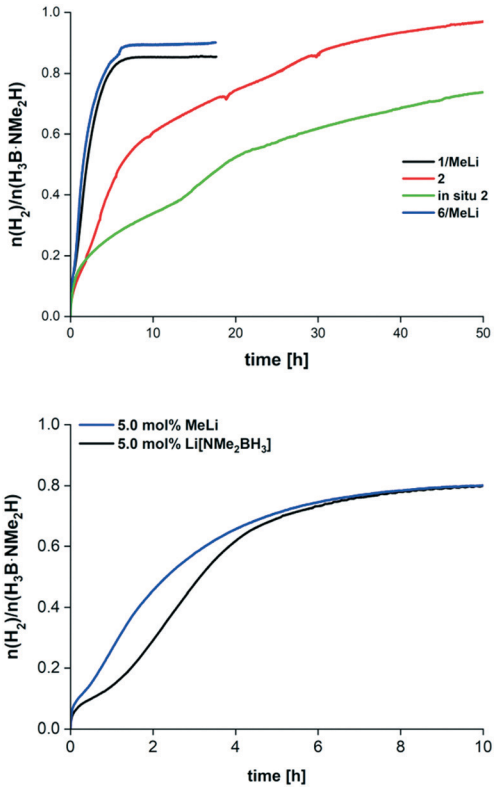


Fig. 5 Top: comparison of volumetric profiles of dehydrocoupling of $H_3B{\cdot}NMe_2H$ using alternative precatalysts and activation protocols (2.5 mol% precatalyst, toluene, 25 °C). Bottom: comparison of volumetric profiles of dehydrocoupling of $H_3B{\cdot}NMe_2H$ using complex **1** and different cocatalysts (2.5 mol% **1**, toluene, 25 °C).

This journal is © The Royal Society of Chemistry 2021

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

period was found for $Li[NMe_2BH_3]$ activated systems (Fig. 5, bottom). We therefore conclude that initial formation of the doubly methylated Zr complex **2** is of limited relevance for the activation of the precatalyst **1**. Instead, reaction *via* a singly methylated complex **6** or activation of $H_3B{\cdot}NMe_2H$ by the cocatalyst MeLi appears to be likely.

NMR spectroscopic monitoring of the dehydrocoupling reaction and catalyst speciation studies. To gain insights into the fate of the Zr catalyst in the post-activation regime of catalysis we have next performed *in situ* NMR experiments. According to ^{11}B NMR spectra (Fig. 6, top) formation of large amounts of the main product $[H_2BNMe_2]_2$ occurs only at a very late stage of the reaction and $H_3B{\cdot}NMe_2BH_2{\cdot}NMe_2H$ is the dominant species at low conversion. Aminoborane $H_2B{=}NMe_2$ and diaminoborane $HB(NMe_2)_2$ are detected at the onset of the reaction, albeit in very low concentration. This distribution of dehydrocoupling products could be rationalised by either initial catalytic formation of $H_3B{\cdot}NMe_2BH_2{\cdot}NMe_2H$ from $H_3B{\cdot}NMe_2H$, followed by further H_2 release to furnish $[H_2BNMe_2]_2$, or catalytic formation of $H_2B{=}NMe_2$ and fast reaction of this species with $H_3B{\cdot}NMe_2H$ to produce $H_3B{\cdot}NMe_2BH_2{\cdot}NMe_2H$ which is then converted into $[H_2BNMe_2]_2$ (*vide supra*). Time-resolved ^1H NMR spectra (Fig. 6, bottom) show the evolution of two singlet resonances at high field (δ −5.12, −5.89 ppm). Notably, these signals resemble those reported before during dehydropolymerisation of $H_3B{\cdot}NMeH_2$ (Fig. 6, bottom).[8]

Based on the previously made mechanistic proposal these signals could indicate the formation of Zr hydride species. Zr amidoborane complexes such as those that were proposed by us before have been described by Roesler and co-workers and should show ^1H NMR resonances at lower field.[28] We have thus next attempted to synthesise a Zr hydride complex starting from the precatalyst **1** and evaluate their role in catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$. While additions of LiH or $LiAlH_4$ to **1** gave ill-defined mixtures, the reaction with $NaBEt_3H$ produced a dark red compound in low yield that according to its ^1H NMR spectrum possesses two different metallocene units and $SiMe_3$ environments. At high field, a signal at −5.09 ppm was detected that a) resembles that observed during catalysis (Fig. 6, bottom) and b) according to its Si satellites and an integration of two with respect to the Cp ligands points to the presence of a CH_2SiMe_3 group. We thus propose that formation of a diamagnetic dinuclear Zr alkynyl–Zr trimethylsilylmethyl complex (**7**) had occurred by cleavage of the allenediide bridging unit (Scheme 3a), possibly with a dinuclear Zr hydride complex as an intermediate. MS analysis shows fragments corresponding to both metallocene-ligand units (*m*/*z*: 537 $[(Cp_2Zr)_2C_2SiMe_3]^+$, 307 $[Cp_2ZrCH_2SiMe_3]^+$, 220 $[Cp_2Zr]^+$) and supports this assignment.

Since the reaction of **1** with $NaBEt_3H$ afforded the presumably catalytically relevant species **7** in low yield, we next evaluated other routes for a more selective preparation of this interesting species. Complexes analogous to **7** have been reported before by Erker and co-workers who have reacted $Cp_2M(Me)(C_2CH_3)$ (M = Zr, Hf) with $Cp_2Zr(\eta^4$-butadiene) to produce related dinuclear complexes $Cp_2M(C_2R)(Me)ZrCp_2$.[29] In a similar, poorly reproducible reaction of light-sensitive $Cp_2Zr(CH_2SiMe_3)(C_2SiMe_3)$ (**8**), prepared by reaction of $Cp_2Zr(Cl)(C_2SiMe_3)$ with $LiCH_2SiMe_3$,[30] with $Cp_2Zr(\eta^4$-butadiene) at 75 °C in benzene we could obtain **7** (Scheme 3b). Details of the synthesis of complex **8** are given in the Experimental section. Attempts to improve this reaction sequence by using the well-known, more convenient alkyne complex $Cp_2Zr(py)$

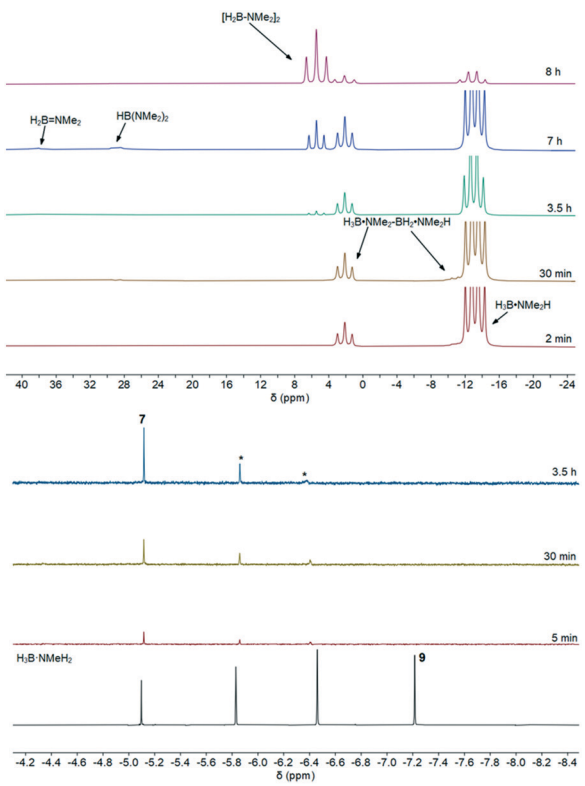
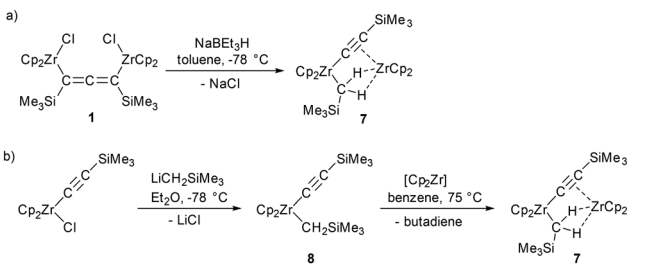


Fig. 6 Top: *in situ* ^{11}B NMR spectra of a dehydrocoupling experiment (2.5 mol% **1**, 5.2 mol% MeLi, 25 °C, toluene-d_8, 96.3 MHz). Centre: high-field region of *in situ* ^1H {^{11}B} NMR spectra of a dehydrocoupling experiment (2.5 mol% **1**, 5.2 mol% MeLi, 25 °C, toluene-d_8, 400.1 MHz). For clarity, the spectrum at t = 8 h is shown at a different scale. Bottom: high-field region of an *in situ* ^1H NMR spectrum of a previously reported dehydropolymerisation experiment (0.02 mmol **1**, 0.05 mmol, 25 °C, benzene-d_6, 300.2 MHz).[8]



Scheme 3 a) Reactions of complex **1** with hydride sources; b) alternative pathway for formation of complex **7** [Cp_2Zr] = $Cp_2Zr(\eta^4$-butadiene).

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

(η^2-$Me_3SiC_2SiMe_3$)[31] (py = pyridine) were not successful. The formation of **7** from **1** during catalytic dehydrocoupling *via* a transient dinuclear Zr dihydride species could involve isomerisation of the allenediide ligand to a propyne unit through a 1,3-H shift and subsequent cleavage of a C–C single bond (Fig. S105†). We have evaluated two different pathways computationally and found an overall exergonic Gibbs free enthalpy change for the transformation of a proposed dinuclear Zr dihydride complex to the experimentally obtained complex **7** ($\Delta_R G^\theta$ = −13.9 kcal mol^{-1}). Although the first transition structure, where a hydride ligand is transferred to the allene unit with an activation barrier of 20.3 kcal mol^{-1} seemed promising, the most reasonable activation barriers of the further proposed reaction paths are much higher (*e.g.* **Int2-TS3** 47.5 kcal mol^{-1}; **Int2_2-TS3**_2 63.7 kcal mol^{-1}, Fig. S105†). These high activation barriers are in accordance with calculated and experimental values for the simplest allene–propyne isomerisation (*e.g.* 65.4 kcal mol^{-1}),[32] which are observed under significantly more drastic conditions.[33–35] Moreover, this can be understood as a reverse reaction of the titanocene mediated alkyne–allene rearrangement found by Mach *et al.*[36] Since the herein described transformation readily takes place at room temperature, formation of **7** seems to follow a different, yet unknown path. Alternatively, complex **7** could be generated during catalysis from a previously proposed, intermediately formed dinuclear Zr bis(amidoborane) complex **3b**. We have evaluated the feasibility of this pathway by reacting the bis(dimethylamide) complex **4** with BH_3·THF, which could potentially yield the desired amidoborane species and ultimately **7** in a consecutive reaction. However, we found that only the borate complex **5** is obtained along with several unidentified metallocene species (Fig. S66†). High field resonances corresponding to **7** or a Zr hydride complex could not be observed.

UV-vis spectroscopic monitoring of catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$. As mentioned above, significant colour changes can be observed during catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$. Singular value decomposition of a set of UV-vis spectra obtained from a reaction using 2.5 mol% **1** and 5.2 mol% MeLi (Fig. S83†) shows minimum three UV-vis active species. In contrast to the substrate free activation of **1** (Fig. 3), in this case the spectrum of the precatalyst **1** could be identified at the beginning of the reaction. It can be suggested that the activation mechanism changes in the presence of $H_3B{\cdot}NMe_2H$, as was shown in stoichiometric studies discussed above. During catalytic reaction, between 0 and 480 min, the evolution of two maxima (385 and 625 nm) takes place, which is in line with the blue-green colour of the reaction system at high conversion (Fig. S83†). The extinction diagram (Fig. S83†) shows that in total no kinetically uniform reaction takes place. The use of MeLi as a cocatalyst causes a lot of side reactions already at the precatalyst activation stage and complicates mechanistic investigations (*vide supra*).

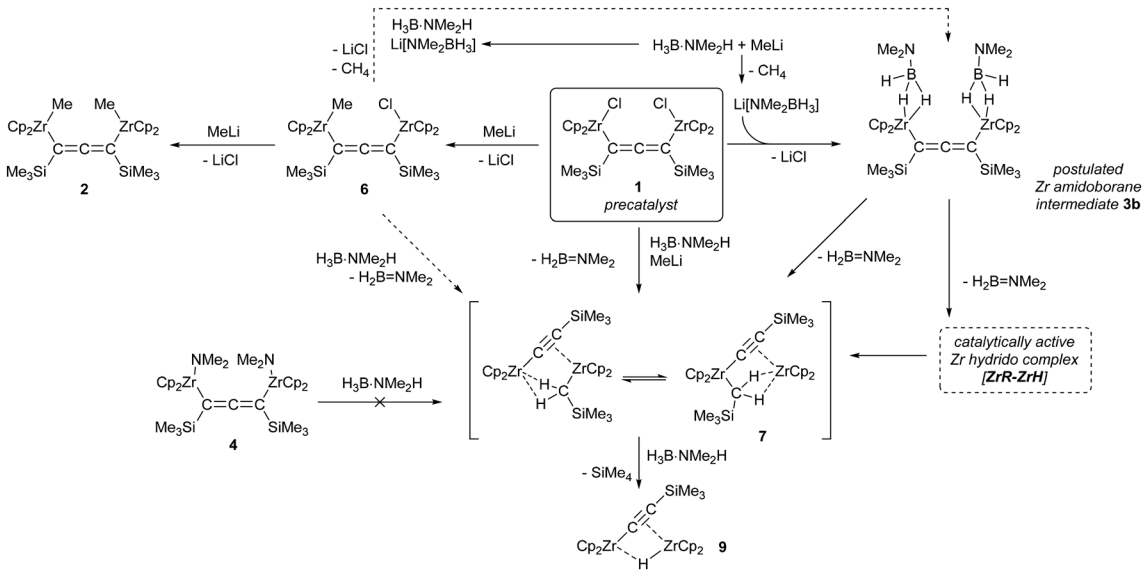
Stoichiometric reactions. Reactions of **1** with MeLi and an equimolar amount of $H_3B{\cdot}NMe_2H$ in benzene-d_6 result in the formation of CH_4 (δ^1H 0.16 ppm) as well as the new dinuclear complex **7** as the dominant species (Fig. S68†). In ^{11}B NMR spectra, besides the presence of dehydrocoupling products $[H_2BNMe_2]_2$ and $HB(NMe_2)_2$, also formation of $[MeHB{=}NMe_2]$ (δ 42.2 ppm) and $[Me_2B{=}NMe_2]$ (δ 45.1 ppm) due to methyl scrambling can be observed (Fig. S69†).[38]

The stoichiometric reaction of **1** with $Li[NMe_2BH_3]$ produced a blue-green mixture of complexes which display the above-mentioned high-field ^{1}H NMR resonances at $-8 < \delta < -5$ ppm (Fig. S62†). While the signal at δ −5.09 ppm is dominant at a ratio **1**/$Li[NMe_2BH_3]$ of 1 : 2, it is absent when four equivalents of $Li[NMe_2BH_3]$ were added. Instead, a new resonance at δ −7.21 ppm occurred. This signal showed no Si satellites, did not sharpen upon B decoupling, and could be associated with further ^{1}H resonances at δ 5.77, 5.76 and 0.10 ppm that were assigned to Cp and $SiMe_3$ groups, respectively. Formation of a Zr hydride complex could thus be assumed. After an extensive literature survey, we assigned the ^{1}H NMR data to a dinuclear Zr(III) hydride complex $Cp_2Zr(C_2SiMe_3)(\mu\text{-}H)ZrCp_2$ (**9**, Scheme 4)[37] in which both Zr centres are bridged by a hydride ligand and a σ, π coordinated alkynyl group. UV-vis analysis of this complex shows the above-mentioned maxima at 385 and 625 nm (Fig. S79†). ^{11}B NMR spectra of stoichiometric reactions of **1** with $Li[NMe_2BH_3]$ showed the formation of cyclic diborazane and the typical dehydrocoupling products. Also μ-aminodiborane, $BH_2(\mu\text{-}Me_2N)(\mu\text{-}H)BH_2$ and $Me_2B{\cdot}NMe_2$ (δ 45.1 ppm)[38] could be observed (Fig. S64†).

For the reaction of Cp_2ZrCl_2 with $H_3B{\cdot}NH_3$ and *n*-BuLi Roesler *et al.* described the formation of Zr σ-N coordinated amidoborane halide and hydride complexes, depending on the stoichiometric ratio of Zr/amine borane.[28] The spectroscopic data presented in this study provide a good estimate for the investigation of possible (dinuclear) Zr amidoborane species derived from precatalyst **1**. Furthermore, protonation of a scandocene alkyl complex was reported to result in σ-B–H coordination of the amidoborane ligand to the metal.[39] For a Rh PONOP pincer system, Weller and co-workers described the formation of a hydride species by reaction of a the chloride precursor with $Li[NMe_2BH_3]$ *via* an amidoborane intermediate.[40] Notably, in this system, σ-B–H coordination is thermodynamically preferred over σ-N coordination by 9.3 kcal mol^{-1}. Variable temperature ^{1}H NMR analysis of the system **1**/$Li[NMe_2BH_3]$ shows that formation of **7** and appearance of the above-described, yet unassigned resonance at δ −6.0 ppm (−5.84 ppm at room temperature) takes place already at low temperature. Apart from a new doublet at δ −5.0 ppm that shows correlation to only one Cp resonance and no $SiMe_3$ resonance no other conclusive features could be identified. ^{11}B NMR analysis at −60 °C revealed the formation of a broad singlet at δ 4 ppm which resolves into a triplet at room temperature. The presence of a terminal or bridging BH_3 group in a putative amidoborane species **3b** is neither in line with these spectroscopic observations, nor with Roesler's

 This journal is © The Royal Society of Chemistry 2021

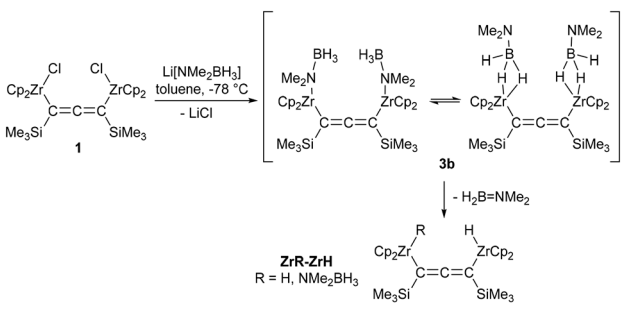
Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.



Scheme 4 Summary of catalyst speciation experiments, including experimentally confirmed and postulated complexes present during dehydrocoupling of $H_3B{\cdot}NMe_2H$ with complex **1**/MeLi.

observation of quartet resonances at much higher field.[28] Detection of $[H_2B{=}NMe_2]_2$ and $H_2B{=}NMe_2$ at room temperature could however point to the presence of such intermediates and formation of a hydride complex by elimination of the aminoborane (Scheme 5).

The reaction of the methyl-chlorido complex **6** with $Li[NMe_2BH_3]$ resulted in the formation of a complex mixture, containing traces of species **7** along with unidentified resonances at δ −5.82 and −6.33 ppm (Fig. S74†). Most interestingly, NMR spectroscopic analysis of a stoichiometric reaction of isolated dimethyl complex **2** with $H_3B{\cdot}NMe_2H$ did not produce any of the high-field resonances corresponding to complexes **7** and **9** (Fig. S72†). Taken together with the comparably slow formation of such Zr dimethyl species, evidenced by NMR and UV-vis experiments discussed above, we thus conclude that formation of this Zr alkyl complex, if at all, only occurs in very small amounts during catalysis. *Activation of the Zr chlorido precatalyst* ***1*** *by reaction with Li[NMe$_2$BH$_3$], formed by MeLi activation of the substrate, is instead more likely to occur.* Formation of monomethyl complex **6** could play a role for formation of amidoborane and hydride species as well.



Scheme 5 Proposed formation of Zr amidoborane and hydride complexes by reaction of complex **1** with $Li[NMe_2BH_3]$.

DFT analysis of the reaction of 1/MeLi with $H_3B{\cdot}NMe_2H$ with a dinuclear Zr hydride or amidoborane ZrR–ZrH species. Based on the previously reported significantly higher catalytic activity of the dinuclear catalyst system **1**/MeLi compared to mononuclear Cp_2ZrCl_2/MeLi,[8] we assumed dehydrogenation of the amine borane and B–N coupling to occur at both Zr centres of the catalyst. Although the total energy gain per suggested catalytic cycle is exergonic (Fig. S103,† $\Delta_R G^\theta = -23.8$ kcal mol^{-1}), we could not calculate any conclusive transition states starting from the proposed dinuclear Zr allenediide bridged intermediates **2** and **3b**. We have investigated formal bimetallic cooperative mechanisms in which an activated amidoborane unit is transferred from one Zr centre to another Zr-bound amidoborane, but only found the formal transfer of the BH_3 group.

Based on the formation of $H_2B{=}NMe_2$ observed in the NMR monitoring experiments (Fig. 6), we next investigated the dehydrogenation of $H_3B{\cdot}NMe_2H$. As suggested by the stoichiometric activation experiments and literature precedent,[28,40] a lithiated amine borane species should be primarily considered as the activating component that should form an amidoborane species **3b** and eventually a catalytically active hydride complex. Computational analysis of dinuclear Zr amidoborane species **3b** shows that similarly as for Weller's Rh system[40] a multicenter Zr–H–B bond is thermodynamically preferred over the Zr–N bonded isomer. Furthermore, this isomer can be considered as a precursor to spectroscopically observed $H_2B{=}NMe_2$ and a catalytically relevant Zr–H species **ZrR–ZrH**. Although speculative, the NMR spectroscopic observation of further unassigned high-field resonances could be regarded as evidence for the presence of such hydride intermediates and the proposed

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

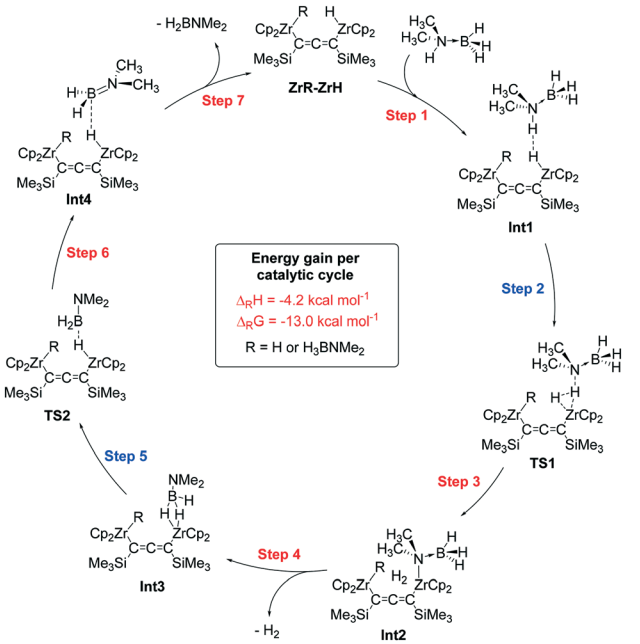


Fig. 7 Proposed mechanism for catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$ with complex **ZrR–ZrH**. Colour code: red – exergonic reaction, blue – endergonic reaction. For details see Table 1.[38]

mechanism of dehydrogenation of $H_3B{\cdot}NMe_2H$ to $H_2B{=}NMe_2$ should include these two features (Fig. 7).[41] The latter presumably reacts metal-free to $[H_2BNMe_2]_2$.

To account for the dinuclear nature of the catalyst and to check the independence of the two Zr centres on the reaction enthalpies, we decided to calculate two formal catalytic cycles, including one (R = H_3BNMe_2) or two Zr–H units (R = H), respectively (Table 1).

Since the overall reaction is the same in both catalytic cycles, they show the same exergonic behaviour of $\Delta_R G^\theta$ = −13.0 kcal mol^{-1}. In the first step, $H_3B{\cdot}NMe_2H$ binds to **ZrR–ZrH** *via* interaction of the NH proton with the hydride ligand. This step is almost thermoneutral in both cases. Starting from these first intermediates (**Int1**), we located transition states (**TS1**) for the formal formation of H_2. Interestingly, this second step shows the highest activation barrier of the catalytic cycle, which is well in line with markedly slower reactions of N-deuterated substrates (*vide supra*).

Table 1 Thermodynamic values (in kcal mol^{-1}) for the proposed catalytic cycle shown in Fig. 7 (ref. 38)

	R = H		R = H_3BNMe_2	
	$\Delta_R H^\theta$	$\Delta_R G^\theta$	$\Delta_R H^\theta$	$\Delta_R G^\theta$
Step 1	−12.1	−0.3	−13.2	−0.9
Step 2	35.3	37.6	26.3	29.7
Step 3	−32.6	−33.7	−24.9	−28.2
Step 4	−2.1	−9.9	−1.3	−8.7
Step 5	10.5	10.1	5.5	5.3
Step 6	−7.1	−8.8	−0.6	−1.3
Step 7	4.0	−7.9	4.1	−8.9

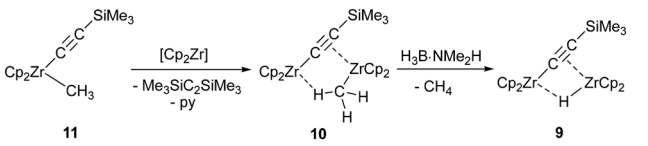
Furthermore, with R = $H_3B{\cdot}NMe_2$ the activation barrier for this step is considerably lowered compared to R = H and the reaction should only occur at room temperature in the first case. Steps 3 and 4 are both exergonic and result in elimination of H_2 and formation of an amidoborane group at the Zr centre (**Int3**). From these structures, release of $H_2B{=}NMe_2$ and recovery of the proposed catalytically active species show only low energy barriers. Dimerisation of $H_2B{=}NMe_2$ to produce $[H_2BNMe_2]_2$ is essentially thermoneutral ($\Delta_R G^\theta$ = −0.5 kcal mol^{-1}) with an energy barrier of 21.8 kcal mol^{-1}.

Evaluation of the role of dinuclear complexes 7, 9, and 10 – bond situation and implications for catalysis

Improved synthesis of the dinuclear hydride 9. For the above-mentioned reasons, the synthesis of complex **9**[37] *via* complex **7** is not convenient (Scheme 3b). We have therefore chosen a slightly different route using the related methyl bridged complex **10** which was prepared by addition of $Cp_2Zr(py)(\eta^2\text{-}Me_3SiC_2SiMe_3)$ to the mononuclear precursor $Cp_2Zr(Me)(C_2SiMe_3)$ (**11**) (Scheme 6). Addition of $H_3B{\cdot}NMe_2H$ to **10** gave the hydride complex **9** in good yield. However, this complex shows a limited stability in solution and decomposes rapidly upon prolonged workup, which has been observed before.[37]

To further evaluate the pathways for the formation of new dinuclear complexes **7** and **9** we have next performed a set of control experiments: In dehydrocoupling of $D_3B{\cdot}NMe_2H$ with **1**/MeLi no signal at δ −5.09 ppm was found, indicating that H transfer occurs from the borane moiety to the bridging allenediide ligand of **1**. Also, addition of $H_3B{\cdot}NMe_2H$ to **7** gave blue-green **9** along with $SiMe_4$ and $[H_2BNMe_2]_2$ (Fig. S30†). Hydride transfer from B–H to Zr–H was confirmed by using $D_3B{\cdot}NMe_2H$, which gave a Zr complex **9-D** that showed no resonance at δ −7.21 ppm. Notably, full conversion of **7** into **9** (Scheme 4) required a large excess of amine borane or performing the reaction in a closed flask to allow buildup of H_2 pressure. In line with this, pressurising a J. Young NMR tube containing **10** with H_2 also slowly furnished **9**.

Discussion of the bond situation in dinuclear complexes 7, 9, and 10. Considering the synthesis sequences, we used for complexes **7** and **10** (Schemes 3b and 6, respectively), two different bond situations are conceivable. The formal oxidation state +4 of the precursors **8** and **11** could be preserved and a Zr(II) centre could be established *via* a π coordination at the alkynyl as well as the alkyl bridge.



Scheme 6 Alternative synthesis of complex **9** [Cp_2Zr] = $Cp_2Zr(py)(\eta^2\text{-}Me_3SiC_2SiMe_3)$.

This journal is © The Royal Society of Chemistry 2021

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

Alternatively, ligand transfer from the Zr(IV) precursor to the Zr(II) fragment could result in redox-comproportionation to yield a dinuclear Zr(III)–Zr(III) complex with ligand assisted antiferromagnetic coupling, making such species readily observable by NMR spectroscopy (Fig. S106 and S107†). The diamagnetic nature of these dinuclear systems was corroborated using EPR spectroscopy which gave no resonances, neither for the isolated complexes **7**, **9**, or **10**, nor for the parent catalytic system **1**/MeLi. Similar group migrations have been described for bis-alkynyl complexes[42] and can proceed very rapidly, making it difficult to distinguish between the two situations described above. To evaluate the bond situation in complexes **7** and **10**, we first calculated the Gibbs free reaction enthalpies for both binding modes as well as the transition states for the respective migration of an alkynyl or methyl group. Interestingly, the two systems differ significantly: whereas in **7** the mixed-valent Zr(IV/II) state is thermodynamically favoured, the Zr(III/III) motif is slightly preferred in **10**, but energetically almost identical (Fig. S106 and S107†). The activation barriers of alkynyl migration are significantly higher ($\Delta_R G^\theta$ = 24.3 (**7**): 28.6 (**10**) kcal mol^{-1}) compared to those of methyl units ($\Delta_R G^\theta$ = 5.3 (**7**) : 1.3 (**10**) kcal mol^{-1}), suggesting interconversion of the two redox isomers *via* alkyl migration. In ^{1}H–^{13}C HMBC NMR experiments **7** shows a correlation of the methylene protons with both carbon atoms of the alkynyl ligand. In contrast, **10** does not show this correlation, which is in line with the different connectivity (Fig. S25 and S37†). SC-XRD analyses unambiguously confirm complex **7** as the Zr(IV/II) isomer and **10** as the Zr(III/III) isomer in the solid state (Fig. 8).

Although the C–C distances of the alkynyl units in **7** and **10** are in the same range (**7**: C2–C3 1.292(8); **10**: C2–C3 1.277(2) Å), which is similar to the values in the dinuclear alkynyl complex $[Cp_2ZrC_2SiMe_3]_2$ (1.249(7)/1.260(7) Å),[43] the distances of the σ-bonded C atom to the respective Zr atom differ significantly (**7**: C2–Zr1 2.010(5); **10**: C2–Zr2 2.124(2) Å). In both complexes, Zr–C(methylene/methyl) distances (**7**: Zr1–C1 2.328(8)/2.903(8); **10**: Zr–C1 2.474(2)/2.520(2) Å) are longer compared to Cp_2ZrMe_2 (2.280(5)/2.273(5) Å).[44] In line with the assumed different structural motifs and formal oxidation states, these are significantly different in **7** and **10**, which should have consequences for the performance of such species in catalytic dehydrocoupling.

Catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$ using dinuclear complexes 7, 9, und 10. As both dinuclear complexes **7** and **9** can be independently prepared and isolated, we have next used these species as precatalysts for dehydrocoupling of $H_3B{\cdot}NMe_2H$ without using MeLi. Under the same reaction conditions as employed before, *i.e.* 2.5 mol% catalyst in toluene with $[H_3B{\cdot}NMe_2H]$ = 1.33 M in an isobaric system, complex **7** only produces approximately 0.2 equivalents of H_2 after 140 hours (Fig. 9). This behaviour is in line with the above-described finding that **7** reacts very slowly with $H_3B{\cdot}NMe_2H$ at ambient conditions. Thus, this species shows no significant contribution to the macroscopically observed catalytic activity of the parent **1**/MeLi system.

In contrast, reactions with the crude material obtained for the hydride complex **9** (approx. 90% purity) produced 0.8 equivalents of H_2 within approximately two days (Fig. 9). Although this reaction is much slower than that of **1**/MeLi, several aspects should be noted: (i) no induction period is observed, (ii) after comparably rapid H_2 evolution at the onset of the reaction, a zero-order type regime follows up to the end of the reaction, (iii) the system shows an intense blue-green colour throughout the course of the reaction, comparable to the final stage of the catalytic dehydrocoupling reaction using **1**/MeLi. After workup of the reaction solution after decease of activity no **9** could be detected by ^{1}H NMR, indicating that this Zr(III) hydride complex is highly sensitive. ^{11}B NMR analysis of the reaction solution at maximum conversion shows traces of residual $H_3B{\cdot}NMe_2H$ as well as $[H_2BNMe_2]_2$ as main product (Fig. S55†).

As we show in Scheme 6, complex **9** can be obtained by reaction of methyl bridged complex **10** with excess $H_3B{\cdot}NMe_2H$. It is thus reasonable to use the latter as a stable single-component precursor to the catalytically active dinuclear Zr monohydrido complex **9**. Monitoring of a catalytic reaction using 2.5 mol% **10** shows that this reaction shows a pronounced induction period, possibly due to slow formation of **9**, that is followed by steady H_2 evolution and close to full

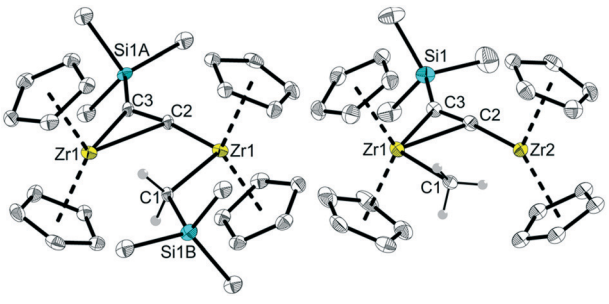


Fig. 8 Molecular structure of complexes **7** and **10**. Thermal ellipsoids correspond to 30% probability. Hydrogen atoms (except those of the Zr bound CH_3 and CH_2 units) as well as one part of disordered moieties in **7** are omitted for clarity.

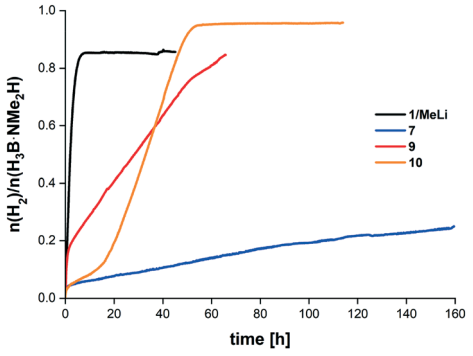


Fig. 9 Comparison of the volumetric curves of dehydrocoupling of $H_3B{\cdot}NMe_2H$ using the parent system **1**/MeLi and dinuclear complexes **7**, **9**, and **10** (2.5 mol% precatalyst, toluene, 25 °C).

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

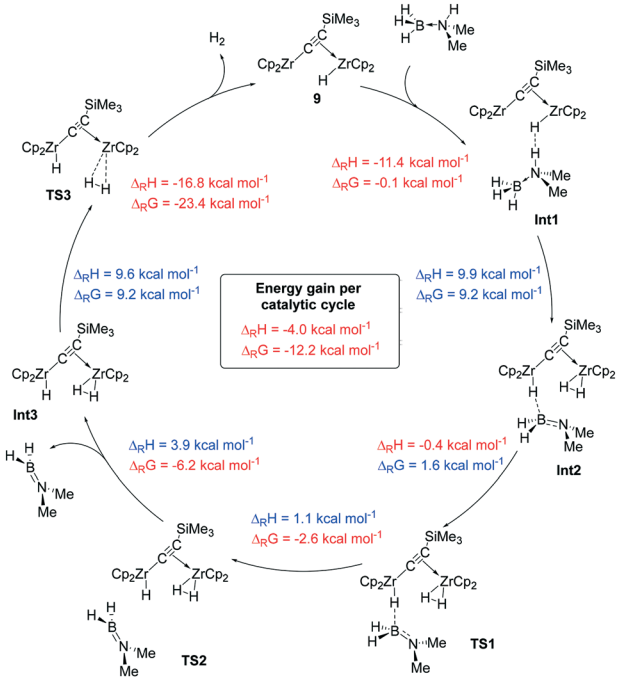


Fig. 10 Proposed mechanism for catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$ with complex **9**. Colour code: red – exergonic reaction, blue – endergonic reaction.

conversion of $H_3B{\cdot}NMe_2H$ (Fig. 9). ^{11}B NMR analysis of the reaction mixture revealed formation of $[H_2BNMe_2]_2$ along with traces of $HB(NMe_2)_2$ (Fig. S57†).

DFT analysis of the catalytic reaction using 9. DFT analysis of the dehydrogenation of $H_3B{\cdot}NMe_2H$ with complex **9** revealed a total energy gain per suggested catalytic cycle of $\Delta_R G^\theta$ = −12.2 kcal mol^{-1} (Fig. 10). We have attempted to model the formation of the spectroscopically observed $[H_2BNMe_2]_2$ *via* the linear diborazane $H_3B{\cdot}NMe_2BH_2{\cdot}NMe_2H$, however, we could not locate possible transition structures that correspond to B–N bond formation at the Zr centres. We thus assume again off-metal coupling to $[H_2BNMe_2]_2$. In a first step $H_3B{\cdot}NMe_2H$ binds to complex **9** *via* interaction of the amine proton with the hydride ligand ($\Delta_R G^\theta$ = −0.1 kcal mol^{-1}) which next forms a dinuclear hydride/dihydrogen complex coordinated by $H_2B{=}NMe_2$ with a moderately low activation barrier (**Int1** → **TS1**: $\Delta_R G^\theta$ = 10.8 kcal mol^{-1}). This is followed by release of $H_2B{=}NMe_2$ and formation of a dinuclear hydride/dihydrogen complex (**TS1** → **Int3**: $\Delta_R G^\theta$ = −8.8 kcal mol^{-1}). From this intermediate the catalytically active species **9** and H_2 are formed *via* **TS3** with a low activation barrier of $\Delta_R G^\theta$ = 9.2 kcal mol^{-1}.

Complexes **9** and **10** can be regarded as dinuclear Zr(III) complexes with antiferromagnetic coupling of both metal centers. M(III) complexes of group 4 metals were discussed as potent catalysts for dehydrocoupling reactions in the past.[5] The herein described complexes are distinct examples for this promising class of complexes as they possess a potentially higher reactivity than M(II) and M(IV) species and offer the possibility of straight-forward NMR tracing of the catalytic process. Furthermore, the herein reported stepwise protocol for the synthesis of dinuclear catalysts (Schemes 3b and 6) should facilitate a systematic investigation of substituent effects at the alkynyl as well as the cyclopentadienyl ligands. Especially the more convenient access to hydride bridged bimetallic complexes such as **9**, compared to the previously reported protocol,[37] allows us to study this class of compounds in detail. These experiments are ongoing in our lab and results will be reported in due course. Moreover, these complexes should hold further potential in related dehydrogenation reactions such as silane dehydrocoupling and dehydrosilylation as described for dinuclear hydride bridged titanium[45] and iron[46] systems.

Conclusions

In summary, we have investigated the mechanism of dehydrocoupling of $H_3B{\cdot}NMe_2H$ using an *in situ* mixture of a dinuclear zirconocene allenediide bridged precatalyst (**1**) and MeLi as a model for the previously described dehydropolymerisation of $H_3B{\cdot}NMeH_2$. We have shown that activation of the Zr complex with MeLi does not instantaneously give the postulated catalytically relevant dimethyl complex **2**. Unexpectedly slow reaction of the organolithium compound with the precatalyst occurs to produce a mixture of Zr monomethyl (**6**) and dimethyl complexes. We thus conclude that the mechanism of precatalyst activation goes beyond the simple reaction with MeLi to produce a dimethyl complex **2** and subsequent protonation by the substrate $H_3B{\cdot}NMe_2H$ to yield a Zr bis(amidoborane) species **3b**. Instead, direct reaction *via* the precatalyst **1** or the monomethyl complex **6** can be regarded as a viable mechanistic alternative. Activation of $H_3B{\cdot}NMe_2H$ and formation of $Li[NMe_2BH_3]$ occurs, the latter activating the precatalyst to produce a Zr amidoborane species and a catalytically active hydride complex. Due to the presence of multiple interconvertible Zr species quantitative kinetic interpretation of this catalytic system is not possible. However, this study offers important insights, *e.g.* regarding the role of free NMe_2H that poisons the early transition metal catalyst, an effect that is contradictory to previous findings for late transition metal-based amine borane dehydrogenation catalysts. We would like to point out that the herein described system represents another example for the commonly applied approach of *in situ* precatalyst activation where intuitively postulated active species are not exclusively responsible for the observed catalytic activity.

Combinations of *in situ* and stoichiometric NMR spectroscopic monitoring experiments indicate degradation of the precatalyst and transient Zr amidoborane or hydride species into at least two dinuclear Zr complexes that possess fragments of the activated allenediide unit of the precatalyst. The identification, independent synthesis and isolation of these unusual Zr(III)/Zr(III) and Zr(II)/Zr(IV) species has allowed for the evaluation of these complexes in catalytic dehydrocoupling of $H_3B{\cdot}NMe_2H$. Although the catalytic activity of these complexes is lower compared to that of the

 This journal is © The Royal Society of Chemistry 2021

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

parent system **1**/MeLi, such compounds could be of relevance for the macroscopically observed catalytic activity of this system in dehydrocoupling and dehydropolymerisation reactions. We have once more shown that dinuclear complexes can be potent catalyst for dehydrocoupling reactions. Initial catalytic tests using the dinuclear Zr(III)/Zr(III) hydride complex **9** suggest that such species can be of interest as well-defined single-component early transition metal catalysts for these and related catalytic applications.

Experimental details

General

All manipulations were carried out in an oxygen- and moisture-free argon atmosphere using standard Schlenk and drybox techniques. The solvents were purified with the Grubbs-type column system "Pure Solv MD-5" and dispensed into Schlenk flasks containing molecular sieves (4 Å) and stored in the refrigerator. Cp_2ZrCl_2 was purchased from MCAT. Complex **1**,[47] $Cp_2Zr(\eta^2\text{-butadiene})$,[48] $H_3B{\cdot}NMe_2BH_2{\cdot}NMe_2H$,[49] $Cp_2Zr(Cl)Me$, $Cp_2Zr(py)(\eta^2\text{-}Me_3SiC_2SiMe_3)$[31] as well as $Cp_2Zr(Cl)CH_3SiMe_3$[50] were prepared following literature procedures. The synthesis of the deuterated amine boranes was done following literature procedures.[51] Complex **2** was synthesised following literature procedures.[8]

NMR spectra were recorded on Bruker AV300 and AV400 spectrometers. ^{1}H and ^{13}C chemical shifts were referenced to the solvent signal: benzene-d_6 (δ_H 7.16, δ_C 128.06), toluene-d_8 (δ_H 2.08, δ_C 20.43).[52]

Raman spectra were recorded on a LabRAM HR 800 Raman Horiba spectrometer equipped with an Olympus BX41 microscope with variable lenses. The samples were excited by different laser sources: 633 nm (17 mW, air cooled), 784 nm laser diode (100 mW, air-cooled) or 473 nm Ar+ laser (20 mW, air-cooled). All measurements were carried out at ambient temperature under argon atmosphere.

IR spectra were recorded on a Bruker Alpha FT-IR, ATR spectrometer which is placed in an Ar drybox. Spectra are not corrected.

MS analysis was done using a Finnigan MAT 95-XP instrument (Thermo-Electron) in CI^+/CI^- mode (isobutene). ESI-MS analysis was done using a Advion expression-CMS instrument.

CHN analysis was done using a Leco Tru Spec elemental analyser. Especially in the case of the bimetallic zirconocene complexes we could not obtain satisfactory elemental analysis in most cases. Despite repeated recrystallisation, repeated measurements with and without oxidiser V_2O_5 and modified furnace temperature, we observed up to 20% less carbon content than calculated/expected. This behaviour might be explained by formation of mixed zirconium–silicon-carbides in the furnace and therefore the carbon content dramatically decreases.[53]

Melting points are uncorrected and were determined in sealed capillaries under Ar atmosphere using a Mettler-Toledo MP 70 device.

X-Ray diffraction data were collected on a Bruker Kappa APEX II Duo and a Stoe IPDSII diffractometer, respectively. The structures were solved by direct methods (SHELXS-97)[54] and refined by full-matrix least-squares procedures on F^2 (SHELXL-2014).[55] CCDC 2046121 (**4**), 2046122 (**6**), 2069778 (**7**) and 2069777 (**10**) contain the supplementary crystallographic data for this paper. XP (Bruker AXS) and Diamond[56] were used for graphical representations.

DFT analysis. First calculations were performed with the pure density functional BP86 (ref. 57) in combination with the LANL2DZ[58] basis set and corresponding effective core potential at Zr and the TZVP[59] basis set on all other atoms. During a series of unsatisfying calculations, we realised that the use of dispersion correction is more feasible for the optimisation of transition state structures as it was described before for the mechanistic elucidation of the dehydrocoupling of $H_3B{\cdot}NMe_2H$ using titanocene catalysts.[60] Therefore we used the well-known hybrid functional DFT method B3LYP[32,61] in combination with basis set def2tzvp[62] and the empirical dispersion correction GD3BJ[63] for all other calculations described herein. If not specifically noted all structures were optimised on this level of theory and vibrational frequencies were also computed, to include zero-point vibrational energies in thermodynamic parameters and to characterise all structures as minima on the potential energy surface. Transition state structures were confirmed by performing intrinsic reaction coordinate (IRC) calculations. In addition to the ESI,† we provide a multi-structure xyz-file including all calculated molecules.

Synthesis of complex 4. Compound **1** (400 mg, 0.57 mmol) was mixed with $LiNMe_2$ (58.6 mg, 1.15 mmol), dissolved in 15 mL of Et_2O and the mixture was stirred overnight. The colour of the reaction mixture changed from red to orange and Et_2O was removed *in vacuo*. The orange solid residue was dissolved in benzene, followed by filtration. After washing with benzene (3 × 2 mL) the solution was concentrated to obtain orange crystals of **4** (326 mg, 82%). Crystals suitable for X-ray diffraction analysis were obtained from a pentane solution. **^{1}H NMR** (25 °C, benzene-d_6, 300 MHz): δ = 0.39 (s, $^2J(^1H, ^{29}Si)$ = 6.2 Hz, 18H, $(SiCH_3)_3$), 2.86 (s, 12H, NMe_2), 5.99 (s, 10H, *Cp*), 6.05 ppm (s, 10H, *Cp*). **^{13}C NMR** (25 °C, benzene-d_6, 100 MHz): δ = 193.6 (C═*C*═C), 111.6 (Cp), 111.3 (Cp), 85.5 (*C*═C═*C*), 51.6 ($N(CH_3)_2$), 3.2 ppm ($SiCH_3$). **^{29}Si-Inept NMR** (25 °C, benzene-d_6, 79.49 MHz): δ = −3.78 ppm ($SiMe_3$). **IR** (ATR, cm^{-1}): ν = 3095 (w), 2815 (vw), 2766 (vw), 1760 (s, C═C═C), 1230 (m), 937 (m), 787 cm^{-1} (s). **Raman** (633 nm, 10 s, 10 scans): ν = 3116 (vw), 3048 (vw), 2954 (vw), 2899 (vw), 2821 (vw), 2771 (vw), 2158 (vw), 1768 (vw), 1443 (vw), 1367 (vw), 1229 (w), 1130 (s), 1069 (w), 943 (m), 845 (vw), 802 (vw), 674 (vw), 625 (vw), 544 (vw), 534 (vw), 352 (vw), 292 (w), 264 (w) cm^{-1}. **MS-CI$^+$** (*isobutene*): *m*/*z* (%): 640 (80) $[(Cp_2Zr)_2(Me_3SiC_3SiMe_3NMe_2)]^+$, 403 (7) $[Cp_2Zr(Me_3SiC_3SiMe_3)]^+$, 263 (15) $[Cp_2ZrNMe_2]^+$, 185 (44) $[Me_3SiC_3SiMe_3]^+$, 109 (100) $[C_3SiMe_3]^+$, 73 (15) $[Si(CH_3)_3]^+$. **Mp**: 159 °C (decomp.). **CHN analysis**: calc (%) for $C_{33}H_{50}N_2Si_2Zr_2$: C, 55.5; H, 7.1; N, 3.9. Found: C, 48.9; H, 6.4; N, 2.6 (best value of four measurements).

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

Synthesis of complex 5. Compound **1** (200 mg, 0.26 mmol) and $NaBH_4$ (20 mg, 0.53 mmol) were dissolved in 15 mL of Et_2O and the mixture was stirred for one week at 40 °C. The orange solution was cannula-filtered, and all volatiles were removed from the filtrate *in vacuo*. The orange foam was dissolved in benzene and was placed in a warm water bath. After slow cooling to room temperature overnight orange crystals of **5** were obtained (152 mg, 76%). It should be noted that complex **5** could not be synthesised in high purity for melting point and CHN analysis. **^{1}H NMR** (25 °C, benzene-d_6, 300 MHz): δ = 0.36 (s, $^2J(^1H, ^{29}Si)$ = 6.4 Hz, 18H, (Si*C*H_3)$_3$), 0.69 (br, 8H, B*H*$_4$), 5.93 (s, 10H, *Cp*), 5.96 ppm (s, 10H, *Cp*). **^{13}C NMR** (25 °C, benzene-d_6, 100 MHz): δ = 186.5 (C═*C*═C), 111.5 (Cp), 111.3 (Cp), 86.2 (*C*═C═*C*), 51.6, 3.7 (Si*C*H_3). **^{29}Si-Inept NMR** (25 °C, benzene-d_6, 79.49 MHz): δ = −5.39 ppm (*Si*Me$_3$). **^{11}B NMR** (25 °C, benzene-d_6, 96.32 MHz): δ = −2.57 (br, *B*H_4). **^{1}H{^{11}B} NMR** (25 °C, benzene-d_6, 300 MHz): δ = 0.69 (br, 8H, B*H*$_4$). **IR** (ATR, cm^{-1}): ν = 3108 (vs), 2951 (s), 2893 (s), 2438 (s), 2390 (s), 2116 (s), 2050 (vs), 1776 (m), 1438 (s), 1319 (s), 1240 (m), 1125 (m), 1065 (s), 968 (s), 828 (m), 729 (w), 674 (m), 620, (m), 515 cm^{-1} (s). **Raman** (633 nm, 4 s, 20 scans): ν = 3127 (vw), 3113 (w), 3101 (vw), 2951 (vw), 2899 (w), 2437 (vw), 2388 (vw), 2196 (vw), 2117 (vw), 1793 (vw), 1437 (vw), 1403 (vw), 1362 (w), 1317 (vw), 1254 (vw), 1231 (vw), 1203 (vw), 1124 (vs), 1067 (vw), 1020 (vw), 1024 (vw), 914 (vw), 907 (vw), 895 (vw), 853 (vw), 843 (vw), 824 (vw), 813 (vw), 805 (vw), 797 (vw), 775 (vw), 676 (vw), 622 (w), 523 (vw), 516 cm^{-1} (vw). **MS-CI$^+$** (*isobutene*): *m*/*z* (%): 655 (2) [M]$^+$, 624 (11) [(Cp$_2$Zr)$_2$(Me$_3$SiC$_3$SiMe$_3$)]$^+$, 415 (45) [Cp$_2$Zr(Me$_3$SiC$_3$SiMe$_3$)BH$_4$]$^+$, 404 (21) [Cp$_2$Zr(Me$_3$SiC$_3$SiMe$_3$)]$^+$, 235 (100) [Cp$_2$ZrBH$_4$]$^+$, 220 (17) [Cp$_2$Zr]$^+$, 185 (2) [Me$_3$SiC$_3$SiMe$_3$]$^+$, 73 (28) [Si(CH$_3$)$_3$]$^+$.

Synthesis of complex 6. Compound **1** (300 mg, 0.43 mmol) was dissolved in benzene (7 mL), MeLi (0.27 μL, 0.43 mmol, 1.6 M in Et_2O) was added slowly and the mixture was stirred overnight at 40 °C. The colour of the reaction mixture changed from red to orange and the solution was filtered off. Benzene was removed *in vacuo*. The orange solid was dissolved in Et_2O and concentrated to obtain **6** as orange crystals (122 mg, 40%). Crystals suitable for X-ray diffraction analysis were obtained as mixed crystals of **6** and **2** from an Et_2O solution. **^{1}H NMR** (25 °C, benzene-d_6, 300 MHz): δ = 0.35 (s, $^2J(^1H, ^{29}Si)$ = 6.4 Hz, 18H, (Si*C*H_3)$_3$), 0.37 (s, $^2J(^1H, ^{29}Si)$ = 6.2 Hz, 18H, (Si*C*H_3)$_3$), 0.42 (s, 6H, –*C*H_3), 5.95 (s, 5H, *Cp*ZrMe), 6.00 (s, 5H, *Cp*ZrMe), 6.05 (d, 5H, *Cp*ZrCl), 6.05 ppm (d, 5H, *Cp*ZrCl). **^{13}C NMR** (25 °C, benzene-d_6, 100 MHz): δ = 175.3 (C═*C*═C), 113.4 (*Cp*ZrCl), 113.2 (*Cp*ZrCl), 111.4 (*Cp*ZrMe), 110.9 (*Cp*ZrMe), 105.7 (C═C═*C*ZrCl), 105.1 (C═C═*C*ZrMe), 36.6 (*C*H_3), 3.3 (Si*C*H_3,) 3.2 ppm (Si*C*H_3). **^{29}Si-Inept NMR** (25 °C, benzene-d_6, 79.49 MHz): δ = −6.42; (*Si*Me$_3$), −7.72; (*Si*Me$_3$).

Synthesis of complex 7

Method A. Compound **1** (300 mg, 0.43 mmol) was dissolved in 20 mL of toluene and the solution was cooled to −78 °C. To the resulting orange solution $NaBEt_3H$ (1.08 mL, 1.08 mmol, 1 M in toluene) was added dropwise over 10 minutes. After stirring overnight, the resulting dark green solution was cannula-filtered and concentrated until precipitation occurred. The green solution was filtered off to obtain red crystals of the product complex. After washing with pentane (3 × 2 mL) complex **7** was received as a dark red solid (78 mg, 26%). **^{1}H NMR** (25 °C, benzene-d_6, 400.13 MHz): δ = −5.09 (s, 2H, C*H*$_2$), 0.33 (s, $^2J(^1H, ^{29}Si)$ = 6.3 Hz, 9H, (Si*C*H_3)$_3$), 0.37 (s, $^2J(^1H, ^{29}Si)$ = 6.5 Hz, 9H, (Si*C*H_3)$_3$), 5.25 (s, 10H, *Cp*), 5.27 ppm (s, 10H, *Cp*). **^{13}C NMR** (25 °C, benzene-d_6, 100.63 MHz): δ = 256.1 (Zr–*C*═C), 179.5 [C═*C*–(SiCH$_3$)$_3$], 104.4 (Cp), 102.4 (Cp), 4.9 (Si*C*H_3), 3.2 (*C*H_2), 1.6 ppm (Si*C*H_3). **^{29}Si-Inept NMR** (25 °C, benzene-d_6, 79.49 MHz): δ = 14.04, (CH$_2$–*Si*Me$_3$), −9.68 ppm (C═C–*Si*Me$_3$). **IR** (ATR, cm^{-1}): ν = 2947 (s), 2887 (vs), 1677 (s), 1436 (s), 1364 (s), 1310 (m), 1234 (m), 1060 (s), 1009 (s), 902 (m), 833 (m), 806 (m), 779 (s), 733 (s), 674 (w), 614 (s), 529 (s), 474 (m), 459 cm^{-1} (m). **Raman** (784 nm, 10 s, 10 scans): ν = 2030 (vw), 1678 (vw), 1438 (vw), 1362 (vw), 1127 (s), 1071 (vw), 844 (vw), 808 (w), 753 (vw), 648 (vw), 594 (vw), 558 (vw) cm^{-1}. **MS-CI$^+$** (*isobutene*): *m*/*z* (%): 626 (100) [M]$^+$, 537 (15) [(Cp$_2$Zr)$_2$C═CSiMe$_3$]$^+$, 307 (19) [Cp$_2$ZrCH$_2$SiMe$_3$]$^+$, 73 (42) [Si(CH$_3$)$_3$]$^+$. **Mp**: 185 °C (decomp.). **CHN analysis**: calc C$_{29}$H$_{40}$Si$_2$Zr$_2$: C, 55.5; H, 6.4. Found: C, 55.8; H, 6.2.

Method B. White solid of complex **8** (16 mg, 0.04 mmol) was mixed with Cp$_2$Zr(η^2-butadiene) (10.8 mg, 0.04 mmol) in a Schlenk flask and dissolved in benzene (0.6 mL). The red solution was stirred at 75 °C for three days. Benzene was removed *in vacuo* and the red solid was dried for 2 h to obtain **7** (12 mg, 75%).

Synthesis of complex 8. *Note: all synthesis steps were performed under exclusion of light.* The white solids of Cp$_2$Zr(Cl)C≡CSiMe$_3$ (64 mg, 0.18 mmol) and LiCH$_2$SiMe$_3$ (19 mg, 0.18 mmol) were placed in a Schlenk flask and dissolved in 10 mL of Et_2O at −78 °C. The colourless solution was stirred at this temperature overnight. The Et_2O was removed *in vacuo* and the white solid was washed with pentane (3 × 0.5 mL) at −78 °C and filtered off. Colourless crystals of **8** could be obtained by storing at −78 °C (53 mg, 83%). **^{1}H NMR** (25 °C, toluene-d_8, 300.2 Hz): δ = 5.91 (s, 10H, *Cp*), 0.64 (s, 2H, C*H*$_2$), 0.24 ppm (s, $^2J(^1H, ^{29}Si)$ = 6.9 Hz, Si(*C*H_3)$_3$), 0.10 ppm (s, $^2J(^1H, ^{29}Si)$ = 6.3 Hz, Si(*C*H_3)$_3$). **^{13}C NMR** (25 °C, toluene-d_8, 100.6 MHz): δ = 160.2 (Zr–*C*≡C), 129.2 (C≡*C*–SiMe$_3$), 110.7 (Cp), 56.6 (Zr*C*H$_2$), 3.5 (Si*Me*$_3$), 0.7 ppm (Si*Me*$_3$). **^{29}Si NMR** (25 °C, toluene-d_8, 79.49 MHz): δ = −1.61 (*Si*Me$_3$), −24.4 ppm (*Si*Me$_3$). **IR** (ATR, cm^{-1}): ν = 3082 (vw), 2948 (w), 2893 (vw), 2026 (w), 1439 (w), 1404 (w), 1329 (w), 1243 (s), 1125 (s), 1015 (m), 834 (m), 799 (s), 715 (m), 678 (m), 601 (w), 413 (w) cm^{-1}. **Raman** (633 nm, 15 s, 10 scans): ν = 3118 (vw), 3085 (vw), 2955 (vw), 2895 (vw), 2839 (vw), 2026 (s), 1440 (vw), 1366 (vw), 1255 (vw), 1125 (s), 1067 (w), 883 (vw), 835 (vw), 812 (vw), 801 (vw), 719 (vw), 698 (vw), 681 (vw), 610 (w), 415 cm^{-1} (w). **MS**: (ESI): *m*/*z* (%): 405 (25) [M]$^+$, 338 (16) [CpZrC≡CSiMe$_3$CH$_2$SiMe$_3$]$^+$, 307 (46) [M–CH$_2$SiMe$_3$]$^+$, 257 (100) [CpZrC≡CSiMe$_3$]$^+$, 229 (96) [CpZrSiMe$_3$]$^+$. **Mp**: 131 °C. **CHN analysis**: calc for C$_{19}$H$_{30}$Si$_2$Zr: C, 56.23; H, 7.45. Found: C, 47.58; H, 2.78 (best value of four measurements).

Synthesis of complex 9

Method A. In a Young NMR tube, complex **7** (25 mg, 0.83 mmol) and H$_3$B·NMe$_2$H (45 mg, 0.75 mmol) were dissolved

 This journal is © The Royal Society of Chemistry 2021

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

in 0.6 mL of benzene-d_6 to furnish a dark red solution. The tube was closed immediately. After one day the colour of the solution had changed from red to blue, indicating the presence of complex **9**, which was confirmed by NMR.

Method B. Complex **10** (133 mg, 0.24 mmol) and $H_3B{\cdot}NMe_2H$ (600 mg, 10.18 mmol) were placed in a Schlenk flask and dissolved in 15 mL of benzene. The mixture was stirred overnight to result in a colour change from red to dark green. All volatiles were removed, and the residue was dried in a warm water bath *in vacuo*. After washing with pentane (4 × 3 mL) the product was obtained as a blue-green solid (97 mg, 72%). Complex **9** must be kept cold to avoid decomposition (decomposition starts immediately when trying to dry or when storing at room temperature). Spectroscopic data are in agreement with previously published data:[37] **^{1}H NMR** (25 °C, benzene-d_6, 400.13 MHz): δ = 5.77 (s, 10H, Cp), 5.76 (s, 10H, Cp), 0.10 (s, $^2J(^1H, ^{29}Si)$ = 6.5 Hz, 9H, Si(CH_3)$_3$), −7.22 ppm (s, 1H, *H*). **^{13}C NMR** (25 °C, benzene-d_6, 100.6 MHz): δ = 283.3 (Zr–*C*≡C), 197.2 (C≡*C*–SiMe$_3$), 105.2 (Cp), 104.3 (Cp), 1.3 ppm (Si*Me*$_3$). **^{29}Si-Inept NMR** (25 °C, benzene-d_6, 79.49 MHz): δ = −13.24 ppm (*Si*Me$_3$).

Synthesis of complex 10. Complex **11** (0.24 g, 0.74 mmol) and $Cp_2Zr(py)(Me_3SiC_2SiMe_3)$ (0.35 g, 0.74 mmol) were placed in a Schlenk flask and dissolved in 50 mL of benzene. The solution was stirred overnight at 60 °C, followed by removal of all volatiles *in vacuo*. The obtained red solid was dissolved in 20 mL of pentane and 0.1 mL of THF. The solution was slowly cooled to −78 °C to yield a red crystalline precipitate. The orange supernatant was filtered off at −78 °C and the red residue was dried *in vacuo* (0.13 g, 33%). **^{1}H NMR** (25 °C, benzene-d_6, 300.2 Hz): δ = 5.19 (s, 10H, Cp); 5.17 (s, 10H, Cp); 0.39 (s, $^1J(^1H, ^{29}Si)$ = 118 Hz, $^2J(^1H, ^{29}Si)$ = 6.5 Hz, 9H, Si(CH_3)$_3$); −2.92 ppm (s, 3H, *Me*). **^{13}C NMR** (25 °C, benzene-d_6, 100.63 MHz): δ = 259.5 (*C*CSiMe$_3$); 180.9 (C*C*SiMe$_3$); 104.5 (Cp); 102.2 (Cp); 2.2 (Zr*Me*); 1.8 ppm (Si*Me*$_3$). **^{29}Si NMR** (25 °C, benzene-d_6, 79.49 MHz): δ = −8.86 ppm ($^2J_{Si,H}$ = 6.5 Hz, SiMe$_3$). **IR** (ATR, cm^{-1}): ν = 445 (m), 542 (m), 612 (m), 683 (s), 744 (vs), 783 (vs), 832 (m), 1011 (m), 1064 (w), 1238 (m), 1435 (w), 1688 (m), 2033 (w), 2948 (w). **Raman** (784, 632, 532, 473 nm): decomposition during measurement with all four available laser sources. **MS-CI$^+$** (*isobutene*): *m*/*z* (%): 554 (100) $[Cp_2Zr(CCSi(CH_3)_3)ZrCp_2CH_3]^+$, 475 (4) [M–Cp–Me]$^+$, 333 (1) [M–ZrCp$_2$]$^+$, 239 (1) [Cp$_2$ZrMe]$^+$, 171 (13) [CpZrMe]$^+$. **Mp**: 149 °C. **CHN analysis**: calc (%) for $C_{26}H_{32}SiZr_2$: C, 56.26; H, 5.81. Found: C, 37.36; H, 5.44 (best value of six measurements).

Synthesis of complex 11. *Note: all synthesis steps were performed under exclusion of light.* Cp$_2$Zr(Me)Cl (2.88 g, 10.59 mmol, 1 eq) and LiC≡CSiMe$_3$ (1.12 g, 12.71 mmol, 1.2 eq) were weighed into a 250 mL Schlenk flask in the glove box. To this white mixture 100 mL of pre-cooled (−78 °C) toluene was added. The resulting white suspension was allowed to slowly warm to room temperature overnight. Filtering off the supernatant and removal of the solvent from the filtrate under vacuum gave a red solid. This solid was repeatedly recrystallised from saturated *n*-hexane solutions until the filter cake was grey. Grey needles of **11** (1.08 g, 32%) were dried under vacuum. **^{1}H NMR** (25 °C, benzene-d_6, 300.2 Hz): δ = 5.88 (s, 10H, *Cp*); 0.26 (s, $^2J(^1H, ^{29}Si)$ = 7.0 Hz, 9H, Si(CH_3)$_3$); 0.18 ppm (s, $^3J(^1H, ^{29}Si)$ = 117.0 Hz, 3H, *Me*). **^{13}C NMR** (25 °C, benzene-d_6, 100.63 MHz): δ = 160.7 (*C*CSiMe$_3$); 127.6 (C*C*SiMe$_3$); 110.9 (Cp); 37.8 (Zr*Me*); 0.9 ppm (Si*Me*$_3$). **^{29}Si NMR** (25 °C, benzene-d_6, 79.49 MHz): δ = −24.3 ppm (*Si*Me$_3$). **IR** (ATR, cm^{-1}): ν = 436 (m), 489 (w), 602 (m), 675 (vs), 740 (s), 795 (vs), 836 (m), 1015 (m), 1244 (m), 1407 (vw), 2023 (vw), 2952 (vw), 3083 (vw). **Raman** (633 nm, 10 s, 20 scans): ν = 303 (w), 490 (vw), 602 (m), 1068 (vw), 1124 (vs), 1365 (w), 2022 (s), 2895 (w), 2960 (vw), 3085 (vw), 3116 (w). **MS-CI$^+$** (*isobutene*): *m*/*z* (%): 414 (13) $[Cp_2Zr(CCSi(CH_3)_3)_2]^+$, 373 (4) [M + iso-butene]$^+$, 333 (100) [M]$^+$, 317 (54) [M–CH$_3$]$^+$, 235 (34) [M–CCSi(CH$_3$)$_3$]$^+$, 220 (8) [Cp$_2$Zr]$^+$, 73 (4) [Si(CH$_3$)$_3$]$^+$. **Mp**: 105 °C (decomp.). **CHN analysis**: calc (%) for $C_{16}H_{22}SiZr$: C, 57.6; H 6.65. Found: C, 56.99; H 6.62.

Synthesis of Li[NMe$_2$·BH$_3$]. $H_3B{\cdot}NMe_2H$ (400 mg, 6.80 mmol) was weighed into a Schlenk flask and dissolved in 15 mL of THF at −78 °C. *n*-BuLi (2.90 mL, 7.50 mmol, 2.50 M in *n*-hexane) was added dropwise and gas evolution was noticed. The reaction was warmed to room temperature and stirred overnight. All volatiles were removed *in vacuo* and the precipitate was washed with pentane (2 × 5 mL). The product was obtained as a white solid (350 mg, 87%). **^{1}H NMR** (25 °C, benzene-d_6, 400.13 MHz): δ = 2.48 (s, 6H, CH_3), 1.36 ppm (br, 3H, BH_3). **^{11}B NMR** (25 °C, benzene-d_6, 96.32 MHz): δ = −14.7 ppm (q, 3H, BH_3).

Synthesis of BH$_2$(μ-Me$_2$N)(μ-H)BH$_2$. $H_3B{\cdot}NMe_2H$ (30 mg, 0.51 mmol) was weighed into a Young NMR tube and dissolved in 0.6 mL of benzene. $H_3B{\cdot}SMe_2$ (48 μL, 0.51 mmol) was added to the colourless solution and the mixture was heated to 80 °C overnight. The product could be observed by NMR spectroscopy. **^{11}B NMR** (25 °C, toluene-d_8, 96.32 MHz): δ = −19.6 ppm (m).

General procedure for dehydrocoupling experiments. $H_3B{\cdot}NMe_2H$ (78.4 mg, 1.33 mmol) and the corresponding precatalyst were weighed in the glovebox and transferred to a three-necked reaction vessel. The $H_3B{\cdot}NMe_2H$ containing reaction vessel was connected to an automatically operating gas buret[6b] under Ar atmosphere. The gas buret was initialised and $H_3B{\cdot}NMe_2H$ and the precatalyst were dissolved in 4 mL of toluene. The MeLi solution (1.6 M in Et$_2$O) was transferred to the reaction vessel and data acquisition was started immediately. After the dehydrocoupling reaction, a gas sample was taken and analysed by GC-TCD. An aliquot was taken from the solution and analysed by ^{11}B NMR spectroscopy (NMR spectra were recorded from *in situ* samples as well as from vacuum-dried samples that were redissolved in NMR solvent).

Numerical rank matrix analysis and singular value decomposition of UV-vis spectra. Numerical rank analyses of absorption matrices are often used to determine the number of linearly independent chemical species. However, this problem can also be solved graphically, using *extinction diagrams*. This not only determines the number of linearly

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

independent partial reactions, but also provides information about the stoichiometric coefficients of the reaction.[64–66] Apart from the first point,[67] the extinction diagram (Fig. S82†) shows straight lines, indicating a kinetically uniform reaction. Thus, it can be assumed that in case of activation of complex **1** with MeLi, the second partial reaction, the reaction of **5** to **2**, was monitored with the experiment (*vide supra*).[68] Further a singular value decomposition (SVD)[69] of the data is used to identify the number of absorbing species. The SVD of a matrix D reads $D = U\Sigma V^T$ with U and V being orthonormal matrices and S being a rectangular up diagonal matrix with the singular values on its diagonal. For a rank-*s*-matrix the first *s* singular values are positive, and the remaining ones are zero. For idealised spectroscopic data based on the Lambert–Beer-model $D = CE^T$ the number of non-zero singular values equals the number of linear independent and absorbing species. For noisy data that follow the Lambert–Beer model approximatively, the same holds for the number of singular values that are larger than a proper threshold. If the signal-to-noise-ratio is well (as for UV–vis-data), then the singular values associated with significant contributions are clearly separated from the ones corresponding to perturbations.[70]

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

We thank our technical and analytical staff, in particular PD Dr. Wolfgang Baumann and Andreas Koch, as well as Dr. Jabor Rabeah for assistance. Financial support by the DFG (project code 419924354) is gratefully acknowledged. F. R. thanks Dr. Jonas Bresien for fruitful discussions[1c,26] about the computational results, the ITMZ at the University of Rostock for access to the Cluster Computer and especially M. Willert for technical support.

Notes and references

1 (*a*) E. M. Leitao, T. Jurca and I. Manners, *Nat. Chem.*, 2013, **5**, 817–829; (*b*) R. L. Melen, *Chem. Soc. Rev.*, 2016, **45**, 775–788; (*c*) T. Beweries and H. Helten, in *Encyclopedia of Inorganic and Bioinorganic Chemistry*, 2020, pp. 1–25, DOI: 10.1002/9781119951438.eibc2717.

2 (*a*) H. Johnson, T. Hooper and A. Weller, in *Synthesis and Application of Organoboron Compounds*, ed. E. Fernández and A. Whiting, Springer International Publishing, 2015, vol. 49, pp. 153–220; (*b*) A. Rossin and M. Peruzzini, *Chem. Rev.*, 2016, **116**, 8848–8872; (*c*) S. Bhunya, T. Malakar, G. Ganguly and A. Paul, *ACS Catal.*, 2016, **6**, 7907–7934; (*d*) D. Han, F. Anke, M. Trose and T. Beweries, *Coord. Chem. Rev.*, 2019, **380**, 260–286.

3 A. Jacobs Elizabeth, A. Fuller, J. Coles Simon, A. Jones Garth, J. Tizzard Graham, A. Wright Joseph and J. Lancaster Simon, *Chem. – Eur. J.*, 2012, **18**, 8647–8658.

4 (*a*) T. J. Clark, C. A. Russell and I. Manners, *J. Am. Chem. Soc.*, 2006, **128**, 9582–9583; (*b*) M. E. Sloan, A. Staubitz, T. J. Clark, C. A. Russell, G. C. Lloyd-Jones and I. Manners, *J. Am. Chem. Soc.*, 2010, **132**, 3831–3841; (*c*) E. A. Jacobs, A. Fuller, S. J. Coles, G. A. Jones, G. J. Tizzard, J. A. Wright and S. J. Lancaster, *Chem. – Eur. J.*, 2012, **18**, 8647–8658.

5 (*a*) D. Pun, E. Lobkovsky and P. J. Chirik, *Chem. Commun.*, 2007, 3297–3299; (*b*) H. Helten, B. Dutta, J. R. Vance, M. E. Sloan, M. F. Haddow, S. Sproules, D. Collison, G. R. Whittell, G. C. Lloyd-Jones and I. Manners, *Angew. Chem., Int. Ed.*, 2013, **52**, 437–440; (*c*) E. A. LaPierre, B. O. Patrick and I. Manners, *J. Am. Chem. Soc.*, 2019, **141**, 20009–20015.

6 (*a*) T. Beweries, S. Hansen, M. Kessler, M. Klahn and U. Rosenthal, *Dalton Trans.*, 2011, **40**, 7689–7692; (*b*) T. Beweries, J. Thomas, M. Klahn, A. Schulz, D. Heller and U. Rosenthal, *ChemCatChem*, 2011, **3**, 1865–1868; (*c*) J. Thomas, M. Klahn, A. Spannenberg and T. Beweries, *Dalton Trans.*, 2013, **42**, 14668–14672.

7 M. Klahn, D. Hollmann, A. Spannenberg, A. Brückner and T. Beweries, *Dalton Trans.*, 2015, **44**, 12103–12111.

8 M. Trose, M. Reiß, F. Reiß, F. Anke, A. Spannenberg, S. Boye, A. Lederer, P. Arndt and T. Beweries, *Dalton Trans.*, 2018, **47**, 12858–12862.

9 (*a*) H. C. Johnson and A. S. Weller, *Angew. Chem., Int. Ed.*, 2015, **54**, 10173–10177; (*b*) A. Kumar, N. A. Beattie, S. D. Pike, S. A. Macgregor and A. S. Weller, *Angew. Chem., Int. Ed.*, 2016, **55**, 6651–6656; (*c*) G. M. Adams, A. L. Colebatch, J. T. Skornia, A. I. McKay, H. C. Johnson, G. C. Lloyd-Jones, S. A. Macgregor, N. A. Beattie and A. S. Weller, *J. Am. Chem. Soc.*, 2018, **140**, 1481–1495.

10 A. L. Colebatch and A. S. Weller, *Chem. – Eur. J.*, 2019, **25**, 1379–1390.

11 Details of the experimental setup were published before in ref. 6*b*.

12 C. D. T. Nielsen and J. Burés, *Chem. Sci.*, 2019, **10**, 348–353.

13 Gas evolution curves showed different kinetic regimes already at less than 10–15% conversion. Analysis of the initial rates of the reaction was therefore not possible.

14 Selected examples: (*a*) C. A. Jaska, K. Temple, A. J. Lough and I. Manners, *J. Am. Chem. Soc.*, 2003, **125**, 9424–9434; (*b*) D. Pun, E. Lobkovsky and P. J. Chirik, *Chem. Commun.*, 2007, 3297–3299; (*c*) A. Friedrich, M. Drees and S. Schneider, *Chem. – Eur. J.*, 2009, **15**, 10339–10342; (*d*) E. M. Titova, E. S. Osipova, A. A. Pavlov, O. A. Filippov, S. V. Safronov, E. S. Shubina and N. V. Belkova, *ACS Catal.*, 2017, **7**, 2325–2333.

15 A. P. M. Robertson, E. M. Leitao, T. Jurca, M. F. Haddow, H. Helten, G. C. Lloyd-Jones and I. Manners, *J. Am. Chem. Soc.*, 2013, **135**, 12670–12683.

16 Details of the computational studies are found in the Experimental section and in the ESI†.

17 (*a*) P. Hasche, J. Haak, F. Anke, C. Kubis, W. Baumann, H.-J. Drexler, H. Jiao and T. Beweries, *Catal. Sci. Technol.*, 2021, DOI: 10.1039/D1CY00124H; (*b*) D. E. Ryan, K. A. Andrea, J. J. Race, T. M. Boyd, G. C. Lloyd-Jones and A. S. Weller, *ACS Catal.*, 2020, **10**, 7443–7448; (*c*) T. M. Boyd, K. A. Andrea, K. Baston, A. Johnson, D. E. Ryan and A. S. Weller,

 This journal is © The Royal Society of Chemistry 2021

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

Chem. Commun., 2020, **56**, 482–485; (*d*) G. M. Adams, D. E. Ryan, N. A. Beattie, A. I. McKay, G. C. Lloyd-Jones and A. S. Weller, *ACS Catal.*, 2019, **9**, 3657–3666; (*e*) F. Anke, S. Boye, A. Spannenberg, A. Lederer, D. Heller and T. Beweries, *Chem. – Eur. J.*, 2020, **26**, 7889–7899; (*f*) A. Glüer, M. Förster, V. R. Celinski, J. Schmedt auf der Günne, M. C. Holthausen and S. Schneider, *ACS Catal.*, 2015, **5**, 7214–7217.

18 E. Hey-Hawkins, *Chem. Rev.*, 1994, **94**, 1661–1717.

19 Equivalents of formed H_2 are referred to n0($H_3B{\cdot}NMe_2H$).

20 Details on the characterisation of this complex, as well as its molecular structure, are depicted in the ESI†.

21 Details on the characterisation of this complex are depicted in the ESI†.

22 In related Zr–BH_4 complexes the BH_4 resonance was either absent or very broad at room temperature. Examples: (*a*) F. Schaper and H.-H. Brintzinger, *Acta Crystallogr., Sect. C: Cryst. Struct. Commun.*, 2000, **56**, e78; (*b*) S. C. P. Joseph, F. G. N. Cloke, C. J. Cardin and P. B. Hitchcock, *Organometallics*, 1995, **14**, 3566–3569; (*c*) J. E. Gozum and G. S. Girolami, *J. Am. Chem. Soc.*, 1991, **113**, 3829–3837.

23 D. Selent and D. Heller, *Catalysis From Principle to Application*, ed. M. Beller, A. Renken and R. van Santen, Wiley-VCH, 2012.

24 H. J. Reich, *Chem. Rev.*, 2013, **113**, 7130–7178.

25 A view of the disordered molecules of the asymmetric unit is shown in the ESI†.

26 Selected examples: (*a*) J. R. Nitschke and T. D. Tilley, *J. Org. Chem.*, 1998, **63**, 3673–3676; (*b*) X. Yan, C. Lai and C. Xi, *Chem. Commun.*, 2009, 6026–6028; (*c*) S. Yu, M. Xiong, X. Xie and Y. Liu, *Angew. Chem., Int. Ed.*, 2014, **53**, 11596–11599.

27 H. Nöth, S. Thomas and M. Schmidt, *Chem. Ber.*, 1996, **129**, 451–458.

28 T. D. Forster, H. M. Tuononen, M. Parvez and R. Roesler, *J. Am. Chem. Soc.*, 2009, **131**, 6689–6691.

29 J. Schottek, G. Erker and R. Fröhlich, *Eur. J. Inorg. Chem.*, 1998, 551–558.

30 Details of the synthesis of complex 9x as well as its molecular structure are given in the ESI†.

31 U. Rosenthal, A. Ohff, W. Baumann, A. Tillack, H. Görls, V. V. Burlakov and V. B. Shur, *Z. Anorg. Allg. Chem.*, 1995, **621**, 77–83.

32 J. H. Kiefer, P. S. Mudipalli, S. S. Sidhu, R. D. Kern, B. S. Jursic, K. Xie and H. Chen, *J. Phys. Chem. A*, 1997, **101**, 4057–4071.

33 Pyrolysis at T > 1000 K: (*a*) H. Hopf, H. Priebe and R. Walsh, *J. Am. Chem. Soc.*, 1980, **102**, 1210–1212; (*b*) C. H. Wu and R. D. Kern, *J. Phys. Chem.*, 1987, **91**, 6291–6296.

34 Photolysis with E = 7–30 eV:K. Alnama, S. Boye-Peronne, S. Douin, F. Innocenti, J. O'Reilly, A.-L. Roche, N. Shafizadeh, L. Zuin and D. Gauyacq, *J. Chem. Phys.*, 2007, **126**, 044304.

35 Treatment at T = 458 K and Fe/Co catalyst:C. P. Khulbe and R. S. Mann, *Can. J. Chem.*, 1978, **56**, 2791–2796.

36 J. Pinkas, R. Gyepes, I. Cisarova, J. Kubista, M. Horacek and K. Mach, *Dalton Trans.*, 2015, **44**, 7276–7291.

37 L. I. Strunkina, M. K. Minacheva, P. V. Petrovskii, Z. S. Klemenkova, B. V. Loshkin, V. V. Burlakov and V. B. Shur, *Russ. Chem. Bull.*, 1997, **46**, 820–823.

38 K. A. Erickson and J. L. Kiplinger, *ACS Catal.*, 2017, **7**, 4276–4280.

39 P. F. Xu and X. Xu, *Organometallics*, 2019, **38**, 3212–3217.

40 E. A. K. Spearing-Ewyn, N. A. Beattie, A. L. Colebatch, A. J. Martinez-Martinez, A. Docker, T. M. Boyd, G. Baillie, R. Reed, S. A. Macgregor and A. S. Weller, *Dalton Trans.*, 2019, **48**, 14724–14736.

41 For the mechanism described here, due to the large number of atoms (>90), the structures were optimized with the def2svpp basis set and in the following single point calculations were performed with the def2tzvp basis set for the calculation of the Gibbs energies.

42 (*a*) G. Erker, W. Frömberg, R. Mynott, B. Gabor and C. Krüger, *Angew. Chem., Int. Ed. Engl.*, 1986, **25**, 463–465; (*b*) G. Erker, W. Fromberg, R. Benn, R. Mynott, K. Angermund and C. Krüger, *Organometallics*, 1989, **8**, 911–920.

43 N. Metzler and H. Nöth, *J. Organomet. Chem.*, 1993, **454**, C5–C7.

44 W. E. Hunter, D. C. Hrncir, R. V. Bynum, R. A. Penttila and J. L. Atwood, *Organometallics*, 1983, **2**, 750–755.

45 C. T. Aitken, J. F. Harrod and E. Samuel, *J. Am. Chem. Soc.*, 1986, **108**, 4059–4066.

46 D. Gasperini, A. K. King, N. T. Coles, M. F. Mahon and R. L. Webster, *ACS Catal.*, 2020, **10**, 6102–6112.

47 F. Reiß, M. Reiß, A. Spannenberg, H. Jiao, W. Baumann, P. Arndt, U. Rosenthal and T. Beweries, *Chem. – Eur. J.*, 2018, **24**, 5667–5674.

48 G. Erker, J. Wicher, K. Engel and C. Krüger, *Chem. Ber.*, 1982, **115**, 3300–3310.

49 C. A. Jaska, K. Temple, A. J. Lough and I. Manners, *J. Am. Chem. Soc.*, 2003, **125**, 9424–9434.

50 S. L. Hart, A. McCamley, P. J. McCormack and P. C. Taylor, *J. Organomet. Chem.*, 1998, **553**, 507–509.

51 N. T. Coles, M. F. Mahon and R. L. Webster, *Organometallics*, 2017, **36**, 2262–2268.

52 G. R. Fulmer, A. J. M. Miller, N. H. Sherden, H. E. Gottlieb, A. Nudelman, B. M. Stoltz, J. E. Bercaw and K. I. Goldberg, *Organometallics*, 2010, **29**, 2176–2179.

53 S. Chen, J. Wang and H. Wang, *Mater. Des.*, 2016, **90**, 84–90.

54 G. M. Sheldrick, *Acta Crystallogr., Sect. A: Found. Crystallogr.*, 2008, **64**, 112–122.

55 G. M. Sheldrick, *Acta Crystallogr., Sect. C: Struct. Chem.*, 2015, **71**, 3–8.

56 *Diamond - Crystal and Molecular Structure Visualization, Crystal Impact - Dr. H. Putz & Dr. K. Brandenburg GbR*, Kreuzherrenstr. 102, 53227 Bonn, Germany, http://www.crystalimpact.com/diamond.

57 (*a*) A. D. Becke, *Phys. Rev. A*, 1988, **38**, 3098–3100; (*b*) J. P. Perdew, *Phys. Rev. B*, 1986, **33**, 8822–8824.

58 (*a*) P. J. Hay and W. R. Wadt, *J. Chem. Phys.*, 1985, **82**, 270–283; (*b*) P. J. Hay and W. R. Wadt, *J. Chem. Phys.*, 1985, **82**, 299–310.

59 A. Schäfer, C. Huber and R. Ahlrichs, *J. Chem. Phys.*, 1994, **100**, 5829–5835.

60 J. Zhu, E.-L. Zins and M. E. Alikhani, *Phys. Chem. Chem. Phys.*, 2018, **20**, 15687–15695.

Published on 13 May 2021. Downloaded by Fudan University on 1/18/2022 2:01:01 AM.

61 (*a*) S. H. Vosko, L. Wilk and M. Nusair, *Can. J. Phys.*, 1980, **58**, 1200–1211; (*b*) C. Lee, W. Yang and R. G. Parr, *Phys. Rev. B*, 1988, **37**, 785–789; (*c*) B. Miehlich, A. Savin, H. Stoll and H. Preuss, *Chem. Phys. Lett.*, 1989, **157**, 200–206; (*d*) A. D. J. Becke, *Chem. Phys.*, 1993, **98**, 5648–5652.

62 F. Weigend and R. Ahlrichs, *Phys. Chem. Chem. Phys.*, 2005, **7**, 3297–3305.

63 S. Grimme, S. Ehrlich and L. Goerigk, *J. Comput. Chem.*, 2011, **32**, 1456–1465.

64 H. Mauser, *Formale Kinetik-Experimentelle Methoden der Physik und der Chemie*, Bertelsmann Universitätsverlag, 1974.

65 J. Polster, Spectrophotometric analysis of simultaneous protolysis equilibria, *Talanta*, 1984, **31**, 113–116.

66 B. Ursin and H. Kleindienst, *Z. Naturforsch., A: Phys., Phys. Chem., Kosmophys.*, 1986, **41**, 989–997.

67 The deviation of the first point can be explained by the fact that it is part of the first partial reaction.

68 This is in direct analogy to the reaction of [Rh(cod)(Cl)]2 with diphosphine ligands, forming a precatalyst of the type [Rh(Cl)(diphosphine)]2. In this case, a stepwise exchange was demonstrated in which the exchange of the first diolefin is considerably faster than that of the second:A. Meißner, A. Preetz, H.-J. Drexler, W. Baumann, A. Spannenberg, A. König and D. Heller, *ChemPlusChem*, 2015, **80**, 169–180.

69 G. H. Golub and C. F. Van Loan, *Matrix computations*, Johns Hopkins University Press, Baltimore, 2012.

70 More details on the reconstruction of the profiles using an SVD can be found in: (*a*) M. Maeder and Y. M. Neuhold, *Practical data analysis in chemistry*, Elsevier, Amsterdam, 2007; (*b*) M. Sawall, H. Schröder, D. Meinhardt and K. Neymeyr, On the ambiguity underlying multivariate curve resolution methods, in *Comprehensive Chemometrics: Chemcial and Biochemical Data Analysis*, Elsevier, 2020, pp. 199–231.

 This journal is © The Royal Society of Chemistry 2021